Eliminating Auxiliary Witnesses in Diophantine Consistency

Polynomial comparison centers, staged index recovery, and a descending-binomial factorial identity

Research prepared for Vladimir Reshetnikov
AI-assisted mathematical construction and independent review

25 October 2026

Abstract

We give three complementary improvements to explicit Diophantine equations for the consistency of ZFC, compared with ProveIt commit `1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61`, including its newly placed packages 127–133. A direct quartic in 314 unrestricted integers has 204 quadratic residuals, 5,721 collected monomials, and coefficient height 128,446. It improves the 324-variable parent in variables, residuals, and support, at the same height. A direct quartic in 233 nonnegative integers has 182 residuals, 7,313 monomials, and height 86,543,666; it dominates both the 234-variable and 233-variable natural parents in the compared metrics. Separate polynomials in ten positive integers have complete 78-operation and 79-operation evaluators, at exact degrees 630 and 468. Their shorter arithmetic circuits trade against larger degrees and retain their parent's large fixed coefficients.

The main new mechanisms are a deterministic polynomial in place of a Chinese-remainder-selected Pell comparison base; recovery of an index from a known divisor before recovering its full denominator; and the identity $F = \lfloor (F^2-1)^t / \binom{F^2-1}{t} \rfloor \iff F = t!$ on the stated positive domain. Complete new ordinary arguments, literal polynomial data, coefficient recipes, exact reconstruction, and independent finite checks are supplied. The direct constructions encode the numerical proof calculus and use neither K215 simulation nor Jones's universal polynomial. The compact branch retains its native calibration. The theorem is an equivalence provable in ZFC, not a proof of consistency or global minimality, and not a new end-to-end formal proof-assistant certificate.

Contents

1 The completed equations and the comparison boundary 3
1.1 What is improved . . . 4
1.2 Why the three mechanisms are useful . . . 4

2 A direct quartic in 314 unrestricted integers 5
2.1 The arithmetic interface and the semantic anchor . . . 5
2.2 An odd, large scale with either sign . . . 6
2.3 A deterministic comparison center . . . 6
2.4 Signed index recovery without a sign guard on the scale . . . 7
2.5 Power semantics and the inherited arithmetic endpoint . . . 8
2.6 Complete witness construction . . . 9
2.7 Exact compilation and checked scope . . . 9

3 A descending-binomial quartic in 233 natural variables 10
3.1 A short factorial fixed point . . . 10
3.2 Descending reciprocal moduli . . . 11
3.3 One cap equation at a semantic index . . . 11
3.4 The six exact powers and the order of recovery . . . 12
3.5 The common root and exact digits . . . 13
3.6 Why horizon 12 is sufficient . . . 14
3.7 From the local arithmetic rows to refutations . . . 14
3.8 Literal compilation and independent review . . . 15

4 An existing Pell denominator removes the artificial scale 15
4.1 Literal polynomial and fixed coefficients . . . 15
4.2 Inherited interfaces and their scope . . . 16
4.3 Root extraction before index or sign semantics . . . 17
4.4 Main folding and transport signs . . . 17
4.5 The new staged index lemma . . . 18
4.6 Projection, source recovery, and positive completeness . . . 18
4.7 Exact cost, degree, and checks . . . 19

5 What the verification proves 20
5.1 Reproduction . . . 21

6 Conclusions and the remaining target 21

A The complete unrestricted-integer residual list 22

B The complete natural-number residual list 32

C The four fixed compact coefficients 41

1 The completed equations and the comparison boundary

Write $\mathbb{N} = \{0, 1, 2, \ldots\}$. There are four completed single polynomials, with three different optimization goals. The domain is part of each specification.

Equation and domain	Variables	Degree	Rows	Terms	Height
Direct, all integers	314	4	204	5,721	128,446
Direct, nonnegative integers	233	4	182	7,313	86,543,666

For the unrestricted-integer endpoint, let $r_0, \ldots, r_{203}$ be the literal residuals in `direct/equation.json`, printed in Appendix A. Its polynomial is

$$P_{\mathbb{Z}}(z_0, \ldots, z_{313}) = \sum_{j=0}^{203} r_j(z_0, \ldots, z_{313})^2. \tag{1}$$

For the natural-number endpoint, let $s_0, \ldots, s_{181}$ be the literal residuals in `natural/equation.json`, printed in Appendix B. Its polynomial is

$$P_{\mathbb{N}}(x_0, \ldots, x_{232}) = \sum_{j=0}^{181} s_j(x_0, \ldots, x_{232})^2. \tag{2}$$

Both lists are fixed finite data. Their indices do not conceal a proof length, an unbounded list, or a variable number of quantified witnesses. Every residual has degree at most two. The complete collected expansions are in the adjacent `expanded_polynomial.json` files.

Compact equation and domain	Variables	Binary operations	Exact degree
`compact/equation78.json`, positive integers	10	78	630
`compact/equation79.json`, positive integers	10	79	468

Every live binary addition, subtraction, and multiplication is counted, including squares and multiplication by fixed numerals. Named registers are abbreviations in a polynomial expression, not extra unknowns. Section 4 gives the complete formulas and proof. The four fixed integers are supplied as literal bytes, with an independently replayed finite reconstruction.

Theorem 1.1 (Consistency calibration)**.** *For the fixed polynomials above, ordinary mathematics formalizable in ZFC proves*

$$\mathsf{ZFC} \vdash \left(\neg\exists \mathbf{z} \in \mathbb{Z}^{314}\ P_{\mathbb{Z}}(\mathbf{z}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}),$$
$$\mathsf{ZFC} \vdash \left(\neg\exists \mathbf{x} \in \mathbb{N}^{233}\ P_{\mathbb{N}}(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

For each of the two compact polynomials P, it also proves

$$\left(\neg\exists \mathbf{u} \in \mathbb{Z}_{>0}^{10}\ P(\mathbf{u}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

Here $\mathrm{Con}(\mathsf{ZFC})$ *is the syntactic absence of a finite refutation in the fixed effective ZFC calculus used by the inherited calibration.*

The constructive assertion proved in each direction is that a polynomial zero exists exactly when a finite refutation exists. No consistency assumption is used in this equivalence. The direct branches compose with the numerical-calculus theorem `KEndpoint.c222_iff_zf_refutation` and the Choice transport `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. Their statements and source identities are retained. The new arithmetic steps are proved here; the named logical calibration is inherited, with its proof boundary recorded explicitly in Section 5.

1.1 What is improved

The repository snapshot contains more than one frontier. In particular, the collection overview had not yet incorporated all of packages 127–133 when those packages were placed. We compared their literal endpoint files, rather than treating the older overview as the latest result. The most relevant direct comparisons are:

Domain	Construction	Variables	Rows	Terms	Height
$\mathbb{Z}$	Package 131	324	208	5,808	128,446
$\mathbb{Z}$	This work	314	204	5,721	128,446
$\mathbb{N}$	Package 129, recommended	234	183	9,019	87,741,134
$\mathbb{N}$	Package 129, denser	233	182	50,762	88,674,690
$\mathbb{N}$	This work	233	182	7,313	86,543,666

The natural polynomial has about 18.9% fewer monomials than the 234-variable parent and about 85.6% fewer than its 233-variable companion. Its height is also smaller than both. The unrestricted polynomial removes ten variables and four residuals, with smaller support and the same height. These are strict improvements in the displayed direct-construction comparisons.

The compact comparison is different. Package 133 supplied 84 operations at degree 436 and 85 operations at degree 328. The new 78/630 and 79/468 endpoints save six operations each but raise degree. Neither compact endpoint is claimed to dominate the older pair in all metrics. Nor do the direct quartics have globally fewest variables: the repository already contains other architectures with many fewer variables, including quartics with very large specialized coefficients. The classical nine-natural-variable and recent ten-integer-variable representation results concern another measure of simplicity [7, 8].

1.2 Why the three mechanisms are useful

An auxiliary witness can cost several final variables after its arithmetic and sign certificates are expanded. The present reductions therefore work at the level of existence proofs before optimizing the resulting polynomial.

1. A Pell comparison base previously chosen by CRT can be an explicit polynomial in an existing Pell trace. Its defining identity supplies both required congruences automatically. This removes a witness choice, not merely a different name for a coordinate.

2. To identify an index, the full error denominator need not be available first. A large known divisor, combined with a strict index bound, suffices. The exact index then recovers the full divisibility. This allows the compact branch to discard its artificial scale.

3. A descending-binomial ratio places the factorial on the other side of a floor. The resulting reciprocal CRT moduli and polynomial interfaces have a smaller collected quartic than the earlier rising binomial construction.

The classical Pell identities, the numerical proof calculus, coefficient encoding, sum-of-squares compilation, semantic-anchor idea, and existing one-row cap idea all retain their attribution. The claims of novelty are relative to the inspected ProveIt sources and concern the stated new lemmas or completed combinations. A literature-wide priority theorem is not asserted.

2 A direct quartic in 314 unrestricted integers

This component uses the numerical C222 proof calculus. It has no K215 simulation and no specialization of a universal polynomial. Its complete ordinary polynomial is

$$P_{314}(z_0, \dots, z_{313}) = \sum_{j=0}^{203} r_j(z)^2 = 0, \qquad z \in \mathbb{Z}^{314}. \tag{3}$$

Every quadratic residual is printed in the appendix and supplied literally in `direct/equation.json`; every collected coefficient is in `direct/expanded_polynomial.json`. The new equation has exact degree four, 5,721 monomials and coefficient height 128,446.

Theorem 2.1. *With the retained ordinary C222-to-ZFC refutation calibration,*

$$\mathsf{ZFC} \vdash \left(\neg\exists z \in \mathbb{Z}^{314}\ P_{314}(z) = 0\right) \iff \mathrm{Con}(\mathsf{ZFC}).$$

Here "retained calibration" identifies previously proved mathematical dependencies; it is not an added consistency hypothesis. The present package does not claim a new end-to-end proof-assistant derivation. The exact parent comparison, independently recomputed from the literal equations, is

Construction	Variables	Rows	Degree	Terms	Height
Package 131 parent	324	208	4	5,808	128,446
This construction	314	204	4	5,721	128,446

Thus the reduction saves ten coordinates, four quadratic rows and 87 collected monomials, without increasing degree or coefficient height. It improves this direct-calculus comparison; other coefficient architectures already use fewer variables.

2.1 The arithmetic interface and the semantic anchor

The pinned parent is package 131 at commit `1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61`. Its accepted factored quadratic source is retained as `direct/baseline/factored.json`, and its complete arithmetic argument as `direct/proofs/parent131.md`. The front has 18 power requests sharing 15 independent Pell pairs. One requested exponent is the derived syntax-label polynomial

$$h = J(A_a, A_b, A_c, A_d).$$

The corresponding pair is literally `index15.X,index15.Y`, and its exponent is `1+add.380`. No new unknown is introduced by writing h.

For the numerical implication constructor

$$I(a, b) = 2(a + b)(a + b + 1) + 4b,$$

J is the fixed sum of the nine axiom conclusions, $a, b, I(a, b)$, the universal quantification constructor and 1. Its original natural circuit has nonnegative coefficients. The retained encoded-column guards give $A_a, A_b, A_c, A_d \geq 1$, so

$$h \geq 16, \qquad h \geq I(A_a, A_b) \geq 2A_a^2. \tag{4}$$

These are raw circuit consequences, prior to interpreting any proof.

The parent has an additional fully indexed Pell cap at its frame index W. We reuse the pair (U, C) already requested at h as that cap. Its principal norm is retained; the old separate cap norm is

deleted. The index-error pool continues to include all fifteen principal pairs, including the anchor. The new frame and cap conditions are

$$W = 2\left(2 + \sum_{a\in\mathcal{A}} a^2\right), \tag{5}$$

$$C^2 - W^2 - 1 - \sum_{i\neq *} y_i^2 \geq 0. \tag{6}$$

Here $\mathcal{A}$ is precisely the retained list of all actual base, exponent and output atoms, including h, with their complete affine restorations. The asterisk marks the anchor. The second condition is one quadratic row using

$$\mathcal{N}(a,b,c) = a^2 + b^2 + c(c+1).$$

This form is nonnegative on $\mathbb{Z}^3$ and represents every natural integer: apply the three-square theorem to $4n+1$, whose three square residues force one odd and two even coordinates.

All other front signs and arithmetic requests are retained. We remove the old sign guard on the scale A and the old sign guard on the auxiliary multiplier, with their three private witnesses each. The ordinary sign argument is given below, rather than borrowing those deleted hypotheses.

2.2 An odd, large scale with either sign

Write $X_n(b), Y_n(b)$ for the integer Pell polynomials, defined by $X_0 = 1, Y_0 = 0$ and

$$X_{n+1} = bX_n + (b^2-1)Y_n, \qquad Y_{n+1} = X_n + bY_n.$$

For negative bases they satisfy

$$X_n(-b) = (-1)^n X_n(b), \qquad Y_n(-b) = (-1)^{n-1} Y_n(b) \quad (n \geq 1).$$

Negative indices use $X_{-n} = X_n$ and $Y_{-n} = -Y_n$.

The original odd-ordinate frame norm remains

$$A^2 - W^3(W+2)(2z+1)^2 = 1. \tag{7}$$

Equation (5) makes W even and $W \geq 516$, hence A is odd. Put $a = |A|$. Classification at center $W+1$ shows that $(a, |W(2z+1)|)$ has a positive Pell index n. As $Y_n(W+1) \equiv n \pmod{W}$, this index is a positive multiple of W. Consequently

$$a \geq X_W(W+1) > W^W, \qquad |A-1| \geq a - 1 > 4W^3. \tag{8}$$

Every semantic atom has absolute value below W. In particular $|k_i| < W$ for every requested index and $16 \leq h < W$.

2.3 A deterministic comparison center

The crucial new local construction replaces a CRT choice by the polynomial

$$\boxed{D = A + (1-A)H^2.} \tag{9}$$

The norms and auxiliary multiplier are

$$U^2 - (A^2-1)C^2 = 1, \tag{10}$$

$$H^2 - (A^2-1)J_c^2 = 1, \tag{11}$$

$$J_c = C^2(2e+1), \qquad e \in \mathbb{Z}. \tag{12}$$

The cap gives $|C| > W$, so C, J_c are nonzero. From the common norm,

$$D = 1 - (A-1)(A^2-1)J_c^2. \tag{13}$$

Since A is odd and $C^2 \mid J_c$, equations (9)–(13) imply

$$D \equiv A \pmod{H}, \qquad D \equiv 1 \pmod{4C}, \qquad |D| > 1. \tag{14}$$

For $A \geq 3$, the second expression is less than -1; for $A \leq -3$ it exceeds 1. Thus neither a sign condition on D nor an adjustable comparison-center parameter is needed.

Retain the comparison relations

$$S^2 - (D^2-1)Q^2 = 1, \qquad S = U + Hg, \qquad Q = h + 4Cq. \tag{15}$$

We now establish the exact absolute anchor index.

Lemma 2.2. *Under the displayed conditions,* $|C| = Y_h(|A|)$.

Proof. Classify the first two norms at $a = |A|$:

$$|U| = X_i(a), \quad |C| = Y_i(a), \quad |H| = X_l(a), \quad |J_c| = Y_l(a),$$

with positive indices i, l. The divisibility $C^2 \mid J_c$ gives $i < l$ and $|C| \mid l$. The latter is the usual squared-ordinate lemma: first $Y_i \mid Y_l$ implies $i \mid l$ by the addition law. Write $l = ik$ and expand the Pell unit power modulo Y_i^2 to obtain

$$Y_l / Y_i \equiv k X_i^{k-1} \pmod{Y_i}.$$

The norm gives $\gcd(X_i, Y_i) = 1$, so $Y_i \mid k$ and hence $Y_i \mid l$.

Classify the comparison pair at $|D|$. Polynomial dependence on the base, $D \equiv A \pmod{H}$, and $S \equiv U \pmod{H}$, with the independent coordinate signs absorbed, imply that its index j satisfies $j \equiv \pm i \pmod{2l}$. This is the parent's abscissa-residue lemma: reduce the unit exponent modulo $2l$, reflect about l, and use strict growth to distinguish the remaining absolute traces. Signed bases contribute only further allowed signs.

Because $D \equiv 1 \pmod{4C}$, one has $Y_j(D) \equiv j \pmod{4C}$. The last comparison congruence gives $j \equiv \pm h \pmod{4C}$. As $|C| \mid l$, these imply $i \equiv \pm h \pmod{2|C|}$. But $1 \leq i \leq |C|$ and $1 \leq h < W < |C|$, so $i = h$. □

The two quadratic rows

$$T - H^2 = 0, \qquad D - A - (1-A)T = 0$$

realize the deterministic center. They replace the old presentation $D = 1 + 4J_c(2d+1)$, $D = A + Hf$. Actual savings are measured after full projection; a local variable subtraction alone would not establish them.

2.4 Signed index recovery without a sign guard on the scale

Each principal norm gives $y_i = \epsilon_i Y_{j_i}(A)$ and $x_i = \pm X_{j_i}(A)$ for some $j_i \geq 0$. For non-anchor pairs the square cap implies $|y_i| < |C|$, hence $j_i < h$; for the anchor $j_i = h$. All are below W. The unchanged pooled row is

$$\sum_{i=1}^{15} (y_i - k_i)^2 = (A-1)q_0, \qquad q_0 \in \mathbb{Z}. \tag{16}$$

The polynomial recurrence gives $Y_j(A) \equiv j \pmod{A-1}$ for every signed A. Thus $|A-1|$ divides

$$E = \sum_{i=1}^{15} (\epsilon_i j_i - k_i)^2.$$

This finite index error satisfies $0 \le E < 4 \cdot 15W^2 < 4W^3 < |A-1|$. Hence $E = 0$, and

$$y_i = Y_{k_i}(A), \qquad x_i = \pm X_{k_i}(A) \tag{17}$$

for all requested indices, including zero or negative ones before the front signs have been decoded. In particular $C = Y_h(A)$ with the correct signed-base convention. We bound E, not the enormous ordinate-error sum in (16).

This signed-base argument credits the repository's earlier unoriented-cap work, retained as `direct/proofs/signed_scale.md`. The present proof rebinds it to the new complete semantic source.

2.5 Power semantics and the inherited arithmetic endpoint

For $k > 0$, $2 \le n < W$ and $|m| < W$, put $T_n = 2An - n^2 - 1$ and

$$L_k = X_k(A) + (n-A)Y_k(A), \qquad R_k = X_k(A) - (n-A)Y_k(A).$$

The identities

$$L_k R_k = 1 + T_n Y_k(A)^2, \qquad L_{k+1} = nL_k + T_n Y_k(A)$$

hold for either sign of A, so $L_k \equiv n^k \pmod{T_n}$. The frame gives $|T_n| > W^W$, larger than both $|m - n^k|$ and $|mn^k + 1|$. The genuine request therefore gives $m = n^k$ on the canonical abscissa branch. The opposite branch would give $mn^k = -1$, impossible in integers. Canonical orientation here need not mean a positive coordinate.

The initial retained rows give $B = r + s + 1 \ge 1$, $q = B + o + 1 \ge 2$, $x \ge 0$ and $t = 16 + h_0^2 \ge 16$. The q^t request first selects the canonical t-pair. Its second request gives $V_0 = (F^2+1)^t$, including the possible base-one case. The ceiling $FB = V_0 + s > 0$ yields $F \ge 1$, then $B > F$ and $q \ge 3$. The cofinal equation $y = V_0(V_0 - 1)$ yields $y \ge F + 1$. Retained encoded-column guards then make every requested index positive, including $K = 2(2 + t + x + y + F)$. The B^K and $(BH_0)^K$ requests recover

$$N = B^K, \qquad q = (BH_0)^K, \qquad H_0 = 1 + \sum \text{the nine positive representatives.}$$

All remaining pairs support requests at q or $q+1$, and the polynomial addition law supplies the derived $t+1$ pair. Thus all 18 power interfaces are genuine. Each retained interval-product guard, together with $r_i + s_i = L_i - 1 \ge 0$, then restores both remainder signs.

At this point the parent's ordinary arithmetic theorem applies to the unchanged front. Its exact proof is retained, with the following chain made explicit. The ceiling gives $V_0 \le BF < B^2 \le q$, so the factorial digit is exact: $B = \binom{F^2+t}{t}$. The parent's fixed-point theorem at $t \ge 16$ yields $F = t!$. The same B equals the product of the pairwise coprime integers $d_i = 1 + F^2/i$, $1 \le i \le t$. Amplified prime-power ranges at $N = B^K$, with the self-sized radix NH_0^K, recover every bounded local rule and derived-label constraint. The Horn-flow multiplicity argument then gives a C222 refutation, even for a supplied multiset containing cycles. The named logical interfaces `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation` give a finite ZFC refutation. The older full formal development is identified rather than recompiled in this release.

2.6 Complete witness construction

Conversely, begin with a finite C222 refutation. Use the parent's cofinal horizon $t = 16 + h_0^2$, truth padding, factorial, beta code and front construction. Fix R_0, K, B, N and choose the nine positive CRT representatives modulo d_i^{2K}. Their common period is N^2. Increase only A_a by a sufficiently large multiple of N^2 so that

$$h \geq 2 \max_{i \neq *} k_i + 1. \tag{18}$$

This is possible by (4): the other varying target is only A_a itself, whereas $h \geq 2A_a^2$. Every local polynomial congruence is preserved at the original stronger precision. The parent has no aggregate upper bound on the representatives.

Only after this lift compute H_0, $q = (BH_0)^K$, all power outputs, digits and finally the doubled frame W. The uniform estimate $J \leq H_0^{2^{22}}$ and $2^{22}(t+1) < K$ preserve all digit bounds; they do not require the representatives to lie below N^2. This ordering avoids circular witness choices.

Choose $A = X_W(W+1) > 0$, which is odd because W is even. The recurrence gives $Y_W(W+1) \equiv W \pmod{2W}$, so

$$z = \frac{Y_W(W+1)/W - 1}{2} \in \mathbb{Z}$$

satisfies the odd frame. Choose all principal pairs canonically, with $U = X_h(A), C = Y_h(A)$. If k_* is the largest other requested index, Pell doubling and growth give

$$Y_{2k_*+1}(A) > (4A - 2)Y_{k_*}(A)^2.$$

Together with (18) and $A > W^W$, this implies $C > W + \sum_{i \neq *} Y_{k_i}(A)^2$. The square-cap slack is therefore nonnegative and has a ternary representation. The pooled quotient exists because $Y_{k_i}(A) \equiv k_i \pmod{A-1}$.

For the auxiliary pair choose $l = hC$ and

$$H = X_l(A), \qquad J_c = Y_l(A) = CY_C(U).$$

The unit-power expansion gives an integer T such that

$$Y_C(U) = CU^{C-1} + (U^2 - 1)T, \qquad \frac{J_c}{C^2} = U^{C-1} + (A^2 - 1)CT.$$

The quotient is a positive odd integer, since A, U are odd. Thus $e = (J_c/C^2 - 1)/2$ supplies (12) for every semantic index h. The even frame extends package 132's even-index odd-quotient construction to this arbitrary anchor index.

Define D by (9) and take $S = X_h(D), Q = Y_h(D)$. These signed integers satisfy the comparison norm, and polynomial base transfer through (14) gives both required congruences. Their unrestricted quotient witnesses therefore exist. Fill the unchanged arithmetic gates, front guards and power quotients. This constructs a zero of the complete new source.

The two existential directions prove the new arithmetic theorem. They may replace old private Pell witnesses; no pointwise bijection fixing those witnesses is asserted.

2.7 Exact compilation and checked scope

The offline rebuild starts at the pinned, literal, already factored parent source. No new Horn factorization is performed. The source certificate lists all twelve modified or removed rows and both appended center rows. An independently written checker reconstructs the whole changed source, including every frame atom and all fifteen pool summands.

Integral monic projection then restores all 216 eliminated coordinates by exact integer polynomials, checks all 94 genuinely absent free coordinates, and verifies the complete retained maps. The final

62 coordinate translations are integral bijections, substituted into every row. Every coefficient of the sum of 204 squares is independently collected. Both the final and parent metrics are computed from literal equations. Three altered-artifact controls are rejected.

These checks import no producer, optimizer or translation helper. The separate rebuild regenerates nine mathematical files byte for byte in a temporary directory. They establish finite algebraic claims; the unbounded Pell, representation, arithmetic and syntactic claims are the ordinary proofs above and in the retained dependencies.

3 A descending-binomial quartic in 233 natural variables

This branch starts from the direct numerical-calculus construction of package 129 [2]. It changes the factorial certificate and the associated reciprocal CRT moduli, combines the semantic anchor with a single cap equation, and uses a checked horizon floor of 12. It retains the Cantor-coded calculus, six exact power requests, eleven modular binomial requests, and the dominant denominator. Its final polynomial has 233 variables, 182 residuals, 7,313 collected monomials, and height 86,543,666.

3.1 A short factorial fixed point

The rising-binomial parent used a ceiling and had to exclude spurious small values before obtaining a strong lower bound for the factorial witness. Reversing the product gives a particularly direct lower bound.

Lemma 3.1 (Descending factorial identity)**.** *Let $t \geq 3$, $F \geq 2$, and $F^2 - 1 \geq t$ be integers. Then*

$$F = \left\lfloor \frac{(F^2-1)^t}{\binom{F^2-1}{t}} \right\rfloor \quad \Longleftrightarrow \quad F = t!.$$

Proof. Put $n = F^2 - 1$, $T = t!$, and $S = t(t-1)/2$. The ratio is

$$H = T \prod_{j=0}^{t-1} \left(1 - \frac{j}{n}\right)^{-1} \geq T.$$

If its floor is F, then $F \geq T$. For $t \geq 3$, $T - 1 > S$; this is immediate at $t = 3$ and persists by induction. Consequently

$$n \geq T^2 - 1 = (T+1)(T-1) > (T+1)S.$$

The elementary product inequality $\prod_j (1 - a_j) \geq 1 - \sum_j a_j$, for $0 \leq a_j \leq 1$, gives

$$T \leq H \leq \frac{Tn}{n-S} = T + \frac{TS}{n-S} < T + 1.$$

Thus the floor is T, so $F = T$. Conversely, if $F = T$, the same displayed inequalities apply and the floor is F. □

The actual polynomial graph makes $F = f_0 + 2$, with $f_0 \in \mathbb{N}$, and evaluates

$$n = (f_0 + 1)(f_0 + 3) = F^2 - 1.$$

This expression has nonnegative coefficients. Its two floor rows are

$$V_0 = BF + r, \qquad B = r + s + 1, \qquad r, s \in \mathbb{N}. \tag{19}$$

After the arithmetic interfaces have recovered $V_0 = n^t$ and $B = \binom{n}{t}$, these rows say exactly $BF \leq V_0 < B(F+1)$. The literal shift $B = b_0 + 2$ ensures $B \geq 2$, so the recovered binomial is nonzero and $n \geq t$. Lemma 3.1 then gives $F = t!$. No factorial, floor, or binomial primitive survives in the final polynomial.

3.2 Descending reciprocal moduli

Lemma 3.2. *Let $F = t!$ with $t \geq 3$. For $1 \leq i \leq t$, put*

$$d_i = \frac{F^2}{i} - 1.$$

These are positive integers greater than one, are pairwise coprime, have no prime divisor at most t, and satisfy

$$\prod_{i=1}^{t} d_i = \binom{F^2 - 1}{t} = B.$$

Proof. Each $i \mid F$, hence F^2/i is a multiple of F and $d_i \equiv -1 \pmod{F}$. Every prime at most t divides F, so no such prime divides d_i. If a prime divided both d_i and d_j, it would divide $id_i - jd_j = j - i$, which is impossible for distinct indices in $\{1, \ldots, t\}$. The product identity is

$$\prod_{i=1}^{t} \frac{F^2 - i}{i} = \frac{(F^2 - 1) \cdots (F^2 - t)}{t!}.$$

□

The encoded local-row index C now uses the polynomial equality

$$C + 1 = F^2 + Bq_C, \qquad q_C \in \mathbb{N}, \tag{20}$$

in place of the parent's $C + F^2 + 1 = Bq_C$. Modulo d_i, $F^2 \equiv i$, so the recovered local index is still $i - 1$. The quotient's natural domain is legitimate in completeness. Indeed, $B > F^2$: with $n = F^2 - 1$, we have $\binom{n}{t} \geq \binom{n}{2} > n + 1$ in the present range. Every positive CRT representative of $F^2 - 1 \pmod{B}$ is therefore at least $F^2 - 1$, and $q_C \geq 0$.

An additional consequence is $d_i \equiv -1 \pmod{4}$, because $4 \mid F$ in the actual $t \geq 12$ construction. Thus each d_i has a prime divisor congruent to 3 modulo 4 with odd multiplicity. This observation is not used to claim another compression in the delivered equation; the preceding elementary CRT facts suffice.

3.3 One cap equation at a semantic index

For $A \geq 2$, write

$$X_j(A) + Y_j(A)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^j.$$

The nonnegative solutions of $x^2 - (A^2 - 1)y^2 = 1$ are exactly these pairs, with unique $j \geq 0$. The elementary recurrences give

$$Y_j(A) \equiv j \pmod{A - 1}, \quad Y_j(A) \geq j, \quad Y_{j+1}(A) > (2A - 1)Y_j(A) \quad (j \geq 1).$$

The complete one-index certificate of the parent is retained for a single pair $(X_*, Y_*) = (X_H(A), Y_H(A))$. The new simultaneous constraint is

$$\boxed{Y_* = H + \sum_{i=1}^{r} (y_i - k_i)^2 + (A - 1)Q, \qquad Q \in \mathbb{N}.} \tag{21}$$

It combines the size cap and the error congruence in one quadratic row. Before the anchor is interpreted, this row already gives $H \leq Y_*$. It therefore supplies the index bound required by the retained one-index certificate. This is a specialization and combination of the earlier one-row cap idea [6] with package 129's semantic anchor, not a priority claim for the general idea of combining a cap and a congruence.

Lemma 3.3. *Assume $W \geq 3$, $2 \leq H < W$, $r \leq W-2$, $0 \leq k_i < W$, $A > W+1$, and $A-1 > W^3$. Let $Y_* = Y_H(A)$, and let each (x_i, y_i) be a nonnegative Pell solution at base A. If (21) holds, then $(x_i, y_i) = (X_{k_i}(A), Y_{k_i}(A))$ for every i. Conversely, for the positive indices in this construction, the canonical pairs satisfy (21) with a natural Q whenever $H > 2\max_i k_i$.*

Proof. Write $y_i = Y_{j_i}(A)$. If $j_i \geq H$, then $y_i \geq Y_* \geq Y_2(A) = 2A > 2W$, so

$$(y_i - k_i)^2 \geq (Y_* - W + 1)^2 > Y_*.$$

This contradicts the nonnegative right side of (21). Therefore $j_i < H < W$. Reducing the row modulo $A-1$ gives divisibility of

$$E = \sum_i (j_i - k_i)^2, \qquad 0 \leq E \leq r(W-1)^2 < W^3 < A - 1.$$

Hence $E = 0$, proving every index equality.

For completeness put $m = \max_i k_i \geq 1$. The identity

$$Y_{2m+1}(A) = Y_{m+1}(A)^2 - Y_m(A)^2 > 4A(A-1)Y_m(A)^2$$

and $H \geq 2m+1$ show

$$Y_H(A) - H - \sum_i (Y_{k_i}(A) - k_i)^2 > [4A(A-1) - r]Y_m(A)^2 - H > 0.$$

This integer is divisible by $A-1$ by the recurrence congruence. Its quotient is the required nonnegative Q. □

The anchor is the existing derived label polynomial $H = J$, which includes the Cantor constructor

$$I(a,b) = 2(a+b)(a+b+1) + 4b, \qquad J(a,b,c,d) \geq I(a,b) \geq 2a^2.$$

As in the parent, the nine independent CRT representatives initially lie in $[1, N^2]$. Increase the encoded a-column by N^2. All local polynomial congruences at precision d_i^{2K} are preserved, and their aggregate remains at most $10N^2$. The new $a > N^2$ gives $H > 2N^4$. Every other required index is at most $2N^2$, so $H > 2\max_i k_i$, as required for the new converse. No extra dominance inequality appears in the polynomial. Soundness uses only the cap lemma's explicit frame hypotheses.

3.4 The six exact powers and the order of recovery

The semantic parameters are

$$t = t_0 + 12, \quad F = f_0 + 2, \quad R = 2 + t + x + y + F, \quad K = 2R, \quad B = b_0 + 2.$$

The six exact requests are

$$V_0 = (F^2-1)^t, \qquad N = B^K, \qquad q = N^K, \qquad D_t = q^t, \quad D_y = q^{y+1}, \quad D_x = q^{x+1}. \tag{22}$$

These are meanings proved for finite polynomial graphs, not operators allowed in $P_{\mathbb{N}}$. The extra denominator $q^{t+1} = qD_t$ uses a multiplication already accounted for in the graph.

The unchanged enlarged growth frame is

$$W = 16 + \sum_{a \in \mathcal{A}} a, \quad Z = z + 1, \quad L^2 - ((W+1)^2 - 1)(WZ)^2 = 1, \quad A = L + \sigma,$$

where $\mathcal{A}$ contains every distinct base, exponent, and proposed output in (22), as well as every modular upper index. All these atoms are natural and less than W. Pell classification at base $W+1$, followed by reduction modulo W, shows that its nonzero index is at least W. Therefore

$$A \geq L \geq X_W(W+1) \geq (W+1)^W, \qquad A-1 > W^3.$$

The actual base A remains free to increase after the lower witness L is chosen. This freedom is needed for the later root congruence.

There are fourteen ordinary index pairs and one fully certified anchor. For ordinary pairs the graph uses consecutive ordinates y, z and a product wire u:

$$u = yz, \qquad y^2 + z^2 - 2Au = 1.$$

Every such pair has an output row of the form $ny = z + R_0$, with $R_0 \geq 0$ and $1 \leq n < A$. Thus $y > 0$, $z < Ay$, and $x = Ay - z > 0$ gives an ordinary Pell pair. The one-row cap authenticates all its indices before interpreting a power or binomial coefficient.

For a proposed power $m = n^k$, put $T_n = 2An - n^2 - 1$ and impose

$$X_k(A) + (n-A)Y_k(A) = m + T_n h_n, \qquad h_n \in \mathbb{N}.$$

Modulo T_n, the left side is n^k. The frame proves $T_n > W^W + W$, $m < W$, and $n^k < W^W$, so both small residues must agree. This proves all six exact powers. The canonical left side is positive and $0 \leq m < T_n$, so their difference cannot be a negative multiple of T_n. Its congruence quotient is therefore natural in completeness. The literal modulus row also excludes a zero power base.

Before using any factorial semantics, $B \geq 2$ gives

$$q = B^{K^2} \geq 2^{4R^2}.$$

With $n = F^2 - 1 < R^2$, every coefficient $\binom{n}{j}$, $0 \leq j \leq t$, is less than $R^{2t} \leq 2^{2Rt} < 2^{2R^2} < q$. Thus the first binomial digit can be decoded before $F = t!$ is known. Its output is B; the floor rows now recover $F = t!$ by Lemma 3.1. There is no circular assumption that a large proposed factorial is already the true factorial.

3.5 The common root and exact digits

The unchanged endpoint rows include

$$y = F + 1 + r_0, \quad x = F + 1 + (y+1)u_0, \quad x = 1 + \big(1 + (t+1)y\big)u_t.$$

They imply $x \geq 2 + (t+1)y > \max(t+1, y)$. Consequently $M = ND_x = Nq^{x+1}$ is a multiple of every needed digit modulus. Put $b = q + 1$ and choose the common-root equation

$$2Ab = b^2 + 1 + Mh_0.$$

It implies

$$\rho_n = X_n(A) + (b-A)Y_n(A) \equiv b^n \pmod{M}.$$

For an exact binomial output $C = \binom{n}{k}$, with $d = q^k$, the literal rows are

$$\rho_n = r + dC + qdh, \qquad d = r + s + 1, \qquad q = C + o + 1.$$

For a range request $N \mid \binom{n}{k}$, they are

$$\rho_n = r + Ndh, \qquad d = r + s + 1.$$

The carry-free binomial theorem gives precisely these interpretations. For completeness, $\rho_n > 0$, while $0 \le r + dC < qd$ for an exact digit and $0 \le r < d \le Nd$ for a range digit. Subtracting the indicated small residue therefore cannot give a negative multiple of the corresponding modulus, so the reconstructed high-digit quotients are nonnegative. Also $N \mid q$, so $\gcd(b, M) = 1$. The linear congruence $2bA \equiv b^2 + 1 \pmod{M}$ is soluble: its only possible common divisor is 2, and then $b^2 + 1$ is even. There are arbitrarily large suitable bases $A \ge L$.

The remaining ten range binomials recover the eight local source columns, the row index, and the derived label sum. Their upper-index and carry bounds are the parent's bounds with the descending factorial input, which is smaller than the former $F^2 + t$. In completeness, the semantic-anchor lift still uses the same $10N^2$ aggregate. For representatives $n \le 10N^2$, relevant lower indices $k \le R - 1$, and $0 \le j \le k$, the old estimate gives $\binom{n}{j} < q$. For the derived label sum, the exact source certificate gives $J \le (n+1)^\ell$, $\ell = 2^{22}$, and $3\ell(t+1) \le 2F \le K$ gives the required stronger bound.

3.6 Why horizon 12 is sufficient

The literal majorant certificate has local exponent $E = 2^{23}$ and label exponent $\ell = 2^{22}$. At the new floor,

$$12! = 479001600, \qquad a = \lfloor \log_2(12!) \rfloor = 28,$$

and direct integer arithmetic verifies

$$aE < 2^a, \qquad 2E \le 2^a, \qquad 3\ell \cdot 13 \le 2 \cdot 12!.$$

These imply the same unbounded inequalities used by the source. In particular, $R^E < 2^R$ starts at $R = 2^a$; the ratio of successive left sides is below 2 for $R \ge 2E$, so it persists for larger integers. The label inequality persists because a factorial grows faster than its linear comparison. Thus the smaller literal horizon changes coefficients without weakening a soundness estimate.

3.7 From the local arithmetic rows to refutations

For clarity, the inherited CRT and logical steps are stated explicitly. If $p^e \mid \binom{a}{k}$, then $a \bmod p^e < k$. For $a \ge k$, this follows from the valuation formula for binomials: if the residue were at least k, the valuation contributions at levels $e, e+1, \dots$ would vanish, leaving valuation at most $e - 1$. The case $a < k$ is immediate. The converse is not assumed.

For each d_i, choose a prime divisor p, let $m = v_p(d_i)$, and work modulo p^{Km}. The index congruence (20) and its range binomial recover $i - 1$. The other range binomials recover the bounded local numerical values. The source majorant bounds its nonnegative local rule polynomial by $R^E < 2^R < 2^K$. Its divisibility by $p^{Km} \ge 2^K$ therefore makes it zero. This produces every local row of the numerical proof calculus.

In completeness use CRT at precision d_i^{2K}, whose common period is $B^{2K} = N^2$. A falling-factorial numerator contains a factor divisible by p^{2Km}, while division by $k!$ loses at most $k - 1 \le K - 1$ powers of p. At least Km powers remain, so all range divisibilities hold. This argument needs only the coprimality and product of the d_i, both proved for the descending moduli above. The semantic lift preserves the same precision.

Finally the local Horn-flow construction uses pairwise coprime weights $w(j) = 1 + jF$. Its beta-coded accumulator starts at $w(1) = F + 1$ and ends at 1. Multiplying the local flow equations and taking prime valuations recovers the multiplicity balance of conclusions and premises. If contradiction label 1 were outside the closure of the selected rules, every rule concluding outside the closure would have an outside premise; the required excess of one outside conclusion is impossible. Conversely a finite proof tree supplies the flow, with padding to any sufficiently large horizon.

This is the retained numerical-calculus equivalence, followed by the two logical interfaces stated in Theorem 1.1.

The order is important: frame and Pell indices; six exact powers; first exact binomial; factorial; remaining binomial ranges; finite local rows; refutation. Completeness chooses a finite proof first, then its bounded semantic data, then the CRT representatives and their lift, and only afterwards the much larger Pell witnesses.

3.8 Literal compilation and independent review

The finite graph exports to 339 natural coordinates and 288 quadratic rows after affine normalization. A further 106 certified eliminations leave the displayed 233 coordinates and 182 rows. Each inverse substitution is an integer polynomial proved nonnegative on the natural orthant; the two restorations with negative coefficients have exact rational sum-of-squares certificates. Expansion gives exactly 7,313 monomials and degree four. All 233 variables occur.

An independent arithmetic review checks the unbounded floor, CRT, cap, and horizon arguments. Its separate coefficient replay verifies the entire expanded dictionary. Bounded controls exercise 2,197 factorial cases, ten reciprocal products, and 2,452 cap tuples; they supplement the proofs and are not used as evidence of universal unsolvability. The files `natural/REVIEW.md`, `natural/INDEPENDENT_REPLAY.json`, and `natural/REPLAY.json` give the exact scopes.

4 An existing Pell denominator removes the artificial scale

The compact branch gives two further fixed equations in ten strictly positive integers. Their exact arithmetic cost and total degree are

$$(78, 630) \qquad \text{and} \qquad (79, 468).$$

Here an operation is one live binary addition, subtraction, or multiplication, including a square or multiplication by a fixed integer. The complete arithmetic circuits are `compact/equation78.json` and `compact/equation79.json`. These are cost–degree tradeoffs against the pinned package 133's $(84, 436)$ and $(85, 328)$ equations. The number of unknowns and the entire fixed coefficient tuple are unchanged.

The new idea is an order of deduction. A large known divisor of a denominator first identifies a Pell index. The identified index then restores divisibility by the whole denominator. Pairwise coprimality of all carrier denominators is unnecessary. This allows us to delete the artificial scale $\mu = q^2 + 1$ and reuse the existing square Y^2. The quotient substitution $Y = q^3 s$, Pell doubling, and the final product-square equation have predecessors in the pinned repository.

4.1 Literal polynomial and fixed coefficients

The only unknowns are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The fixed integers β, K_0, M_C, M_F are the complete directly indexed BFS943 tuple of package 133. They satisfy

$$\beta \geq 31 \text{ odd}, \qquad K_0 > 0, \qquad 0 < M_C < \beta < M_F < 2\beta.$$

The supplied compressed unsigned little-endian payloads fix every byte; `compact/rebuild_coefficients.py` reconstructs all four integers from the unchanged finite tile specification and its 436-character code. All four reconstructed raw hashes agree with the pinned parent. The largest fixed port has 592,884 bits. This is the size of the stored constant, not the height of the fully expanded polynomial.

All following symbols are polynomial abbreviations:

$$q = \beta J + 1, \quad h = q - 1, \quad C = q - F - \alpha, \quad X = wq,$$
$$Y = q^3 s, \quad k = \eta + \zeta, \quad c = kY + \eta, \quad D = Y^2,$$
$$A = Y(X+1) + 1, \quad \Delta = A^2 - 1, \quad H = 4A - 5,$$
$$V_0 = XD, \qquad E = 1 + V_0(V_0+1)k^2,$$
$$R = [h(q-F) + \alpha](q^2-1) + (M_C + qM_F)J,$$
$$e = k - R - 1, \quad B = (K_0 + w)C - F, \quad U = ic^2,$$
$$a = \Delta U^2 E, \quad F_0 = 1 + a, \quad G = EF_0,$$
$$F_* = 1 + 4aF_0, \qquad Q = \Delta(F_* - 1),$$
$$b = c(T+\alpha) - R, \quad V = F_* b - c, \quad K_+ = X(3c - 2X) + 2.$$

Choose $L = V^2$ for the 78-operation equation, or $L = bV$ for the 79-operation equation, and define

$$\begin{aligned} M &= hDH, \\ Z &= hD\big[H(n - LB) + K_+\big] - H(DB + he), \\ R_G &= M^2 G. \end{aligned}$$

The single equation is

$$\boxed{P_L = Q(Z^2 - R_G V^2) + (R_G - Z^2) = 0.} \tag{23}$$

There are no additional quantified variables, nonpolynomial primitives, or implicit conditions in this definition. The actual circuits compute $A-1$, $\Delta = (A-1)((A-1)+2)$, and $H = 4(A-1)-1$.

Theorem 4.1 (Unscaled compact consistency equations)**.** *For either stated choice of L, ordinary arithmetic formalizable in ZFC, composed with the pinned indexed native/refutation calibration, proves*

$$\mathsf{ZFC} \vdash \big(\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{10}\ P_L(\mathbf{x}) = 0\big) \iff \mathrm{Con}(\mathsf{ZFC}).$$

The exact degrees are 630 and 468, and the supplied evaluators have 78 and 79 live binary operations, respectively.

4.2 Inherited interfaces and their scope

The proof uses the following explicitly retained ordinary arithmetic interfaces. They are statements of earlier work, not claims made by a finite coefficient checker.

1. The universal main/auxiliary Pell folding and rank theorems: from the strong norm, $c^2 \mid I$, and the fixed-minus auxiliary residues, they first recover $c = \psi_A(p)$ for an odd positive p, and, once $0 < p, R < c/2$, recover $p = R$.

2. The universal projection and fixed-minus parity theorems: after the required index and size bounds, they give the whole-product projection modulo $4A - 5$ and $R \equiv 3 \pmod 4$.

3. The zero-offset ratio estimate, proved uniformly for every $\lambda \geq 1$, and its half-binomial rounding consequences.

4. The indexed native coefficient calibration and fresh source reconstruction: the resulting outer arithmetic data are equivalent to a finite syntactic ZFC refutation. This branch retains its native/K215 ancestry; it is separate from the direct-calculus quartics.

The exact source files and immutable commit are recorded in `compact/SOURCE_PINS.json`. The new argument below verifies the changed premises and supplies both existential directions. It does not claim a new end-to-end proof-assistant derivation.

4.3 Root extraction before index or sign semantics

For proof notation only put

$$\delta = Q - 1, \quad D_y = QV^2 - 1, \quad z = Z/M, \quad \kappa = K_+/H, \quad \rho = e/D, \quad t = B/h, \quad W = 1 + hL.$$

The identities

$$P_L = \delta Z^2 - M^2 G D_y, \qquad z = n + \kappa - Wt - \rho \tag{24}$$

hold without divisibility assumptions. Positivity of the inputs gives $q \geq 32$, $X \geq q$, $Y \geq q^3$, $k \geq 2$, $c \geq 3$, $A > 4$, $\Delta \geq 3$, $U \geq c^2$, $E > 1$, $F_* > c + 1$, $Q > 1$, and $M > 0$. For every integer b, $V = F_* b - c \neq 0$: at $b = 0$ it is $-c$; otherwise $|V| \geq F_* - c > 1$. Therefore $D_y > 0$.

Identically,

$$F_* = (1 + 2a)^2, \qquad Q = 4\Delta^2 U^2 G.$$

Thus G is coprime to both δ and D_y. A zero of (23) gives $\delta z^2 = G D_y$. For each prime dividing G, its valuation in G is even, because the other two factors are units at that prime and a rational square has even valuation. Hence $G = g^2$ with $g > 0$ integral. Since $F_0 \equiv 1 \pmod{E}$, the positive coprime factors split:

$$E = \tau^2, \qquad F_0 = f_0^2, \qquad g = \tau f_0.$$

Then $y = |z|/g$ is a positive rational satisfying

$$QV^2 - (Q - 1)y^2 = 1. \tag{25}$$

For completeness, the special adjacent form forces y integral. Writing $y = r/m$ in lowest terms gives $m^2 \mid Q - 1$. If $v = |V| > 1$, then $v < y < Qv/(Q - 1)$. The norm-preserving transformation

$$v' = (2Q - 1)v - 2(Q - 1)y, \qquad y' = (2Q - 1)y - 2Qv$$

has an integer first coordinate with $0 < |v'| < v$ and preserves the reduced denominator m of its second coordinate. Reflecting signs and descending reaches $v = 1, y = 1$, so $m = 1$. Consequently

$$z = \varepsilon g y \in \mathbb{Z}, \qquad \varepsilon \in \{-1, 1\}.$$

Set $f = 1 + 2a$ and $I = 2Ug$. Pell doubling gives

$$f^2 - \Delta I^2 = 1, \qquad Q = (\Delta I)^2, \qquad c^2 \mid I, \qquad V = f^2[c(T + \alpha) - R] - c. \tag{26}$$

No residue involving a scale μ has been used.

4.4 Main folding and transport signs

The folding interface applies to (25) and (26), since $c \leq I$ and $V \equiv -c \pmod{f}$. It gives $c = \psi_A(p)$ for odd $p \geq 1$. The first norm yields $k = 2\psi_{2V_0+1}(n_0)$ for $n_0 \geq 1$. If $n_0 = 1$, then $k = 2$, $\eta = \zeta = 1$, and $c = 2Y + 1$ lies strictly between the first two main ordinates 1 and $2A$, impossible. Therefore

$$k \geq 8V_0 + 4, \qquad c > kY \geq k > 8XY^2 \geq 8q^7. \tag{27}$$

This replaces the earlier scale-based $8q^5$ bound by a stronger one. It proves $c > X$ and $K_+ > 0$. The unconditional mask estimate $R < q^4 + (\alpha + 2)q^2$, together with (27), gives $b > c$ before any sign is assigned to B. Thus $V > 0$, $\kappa > 0$, and

$$b - e = c(T + \alpha) - k + 1 > 0, \qquad \rho < b/D. \tag{28}$$

For $L = V^2$, the identity

$$a^2 - g^2 = E[\Delta U^2 E(\Delta U^2 - 1) - 1] > 0$$

gives $g < a$, whence $V > (F_* - 1)b > ab > 3g$. Since $y < 3V/2$ by (25), $gy < L/2$.

For $L = bV$, the same fifth-index argument as in package 133 gives $b > 4g$. In detail, $c \geq 3$ makes $p \geq 3$, so $c \geq 4\Delta + 3$ and $V > 4Q$. Put $S = \Delta I$, so $Q = S^2 \geq 4$. Classification of (25) gives $SV = \chi_S(j)$ with odd j. The first two odd indices give $V = 1, 4Q - 3$, both excluded. Hence

$$V \geq 16Q^2 - 20Q + 5 > 8Q^2, \qquad b > 4\Delta^2(F_* - 1) = 4\Delta^3 I^2 > 4g.$$

It follows that $gy < 3bV/8 < L/2$ here as well.

If $B \leq -1$, then $Wt \leq -L - 1/h$. Equations (24) and (28) imply

$$n = \varepsilon gy - \kappa + Wt + \rho < L/2 - L + b/D < 0,$$

because $V > b > 1$ and $D > 2$. This contradicts $n > 0$. Therefore $B \geq 0$ and $C > 0$. The outer mask bounds now give $3q < R < q^4 - 2$. The rank interface has all its premises: $c^2 \mid I$, $c > 2R$, $c > 2p$, $V \equiv -c \pmod f$, and $V \equiv -R \pmod c$. It gives $p = R$; the unchanged mask parity then yields even q and odd J.

4.5 The new staged index lemma

Lemma 4.2 (Partial divisibility recovers the full denominator)**.** *At every positive zero of* (23)*,*

$$n_0 = (R + 1)/2 \qquad \textit{and} \qquad Y^2 \mid e.$$

Neither $\gcd(Y, q - 1) = 1$ *nor integrality of* e/Y^2 *is needed as a premise.*

Proof. The integral carrier in (24), after clearing hH, implies $Y^2 \mid hHe$. Since $Y = q^3 s$, one has $q^6 \mid Y^2$. Moreover $h = q - 1$ and $H = 4Y(X + 1) - 1 \equiv -1 \pmod q$, so $\gcd(q^6, hH) = 1$. Consequently $q^6 \mid e$.

Let $P_0 = 2XY^2 + 1$. The base-one Pell recurrence gives

$$\psi_{P_0}(n_0) \equiv n_0 \pmod{P_0 - 1}, \qquad k \equiv 2n_0 \pmod{4XY^2}.$$

It follows that $q^6 \mid 2n_0 - R - 1$. But $P_0 - A = Y(2XY - X - 1) > 0$. If $n_0 \geq R$, monotonicity in the base and index gives $k/2 = \psi_{P_0}(n_0) \geq \psi_A(R) = c$, contradicting $c = kY + \eta > k$. Thus $1 \leq n_0 < R$, and

$$|2n_0 - R - 1| < R < q^4 < q^6.$$

The divisible integer must be zero. Returning to the *full* first-Pell congruence now gives $4XY^2 \mid k - R - 1 = e$, and in particular $Y^2 \mid e$. □

The modulus q^6 is used only in this proof. It introduces no new polynomial register. The whole denominator is recovered after the index, which is the step that removes the artificial scale.

4.6 Projection, source recovery, and positive completeness

Write $R = 2r + 1$, with $r \geq 48$. The retained zero-offset ratio lemma holds for every $\lambda \geq 1$ with $A_\lambda = \lambda Y(X + 1) + 1$ and $P_\lambda = 2X\lambda^2 Y^2 + 1$. At $\lambda = 1$ it gives

$$\xi < \frac{\psi_A(2r + 1)}{\psi_{P_0}(r + 1)} < \xi \left(1 + \frac{1}{Y(X + 1)}\right)^{2r}, \qquad \xi = \frac{(X + 1)^{2r}}{X^r}.$$

The lemma explicitly includes this value of λ. Its lower comparison follows from

$$(2A_\lambda - 1)^2 X - (X + 1)^2 (2P_\lambda) = 4\lambda Y(X + 1)X + X - 2(X + 1)^2 > 0;$$

the upper comparison uses $A_\lambda \le \lambda(Y(X+1)+1)$. Together with $Y < c/k < Y+1$, it yields $Y > X^r/3$ and $X^{r+1} < 3Y(X+1) < H$.

Now ρ is integral, so the carrier gives $H \mid hK_+$. The identity

$$K_+ = HXc - 2K_{\mathrm{quad}}, \qquad K_{\mathrm{quad}} = X[X + 2(A-2)c] - 1$$

allows cancellation of two through odd H. The universal projection therefore implies

$$H \mid h(X - 2^R)(2^R X + 1).$$

The retained whole-product bound applies with the same $X \ge q \ge 32$ and $r \ge 48$; its old multiplier $q^3 h$ was larger. Thus the entire product has absolute value below $X^{r+1} < H$, and $X = 2^R$. No cancellation of a possibly nonunit factor modulo a composite H is assumed. It follows that κ is integral. The remaining carrier gives $h \mid WB$, hence $h \mid B$ because $W \equiv 1 \pmod h$. The quotient $t \ge 0$ is integral and $q^3 \mid Y$ is literal. Fixed-minus parity and exact half-binomial rounding now reconstruct the earlier source with its *fresh* base $Y(X+1)+2$. The pinned native calibration yields a finite ZFC refutation.

Conversely, a refutation supplies canonical outer data with $X = 2^R$, $R \equiv 3 \pmod 4$, $r \ge 48$, $q^3 \mid Y$, and $t = B/h \ge 0$, together with the half-binomial identity. Set $s = Y/q^3 > 0$. At the new unscaled bases take

$$c = \psi_A(R), \quad k = 2\psi_{P_0}(r+1), \quad \tau = \chi_{P_0}(r+1), \quad \eta = c - kY, \quad \zeta = k - \eta.$$

The uniform ratio bound and the same rounding estimates prove $0 < \eta < k$, so both new coordinates are positive. The base-one recurrence gives e a positive multiple of $4XY^2$, so $\rho \ge 1$; projection gives a positive integral κ.

Choose a seed Pell pair (f_0, I_0) at a positive multiple of the finite matrix period modulo $c^2\tau$, and set $i = I_0/(c^2\tau) > 0$. Then $U = I_0/\tau$, $F_0 = f_0^2$, and $G = (\tau f_0)^2$. Double this pair:

$$f = 2f_0^2 - 1, \qquad I = 2f_0 I_0 = 2Ug.$$

The retained auxiliary-completeness theorem accepts this strong pair without a restriction on the parity of its index. It supplies $T' = T + \alpha > \alpha$ and positive V, y satisfying the stated auxiliary equation, so $T > 0$. Finally choose

$$n = gy + Wt + \rho - \kappa.$$

Every term is integral. Since $f_0 > (A-1)I_0 > Xc > \kappa$, one has $g > \kappa$; hence $n > 0$. The resulting carrier is exactly $z = gy$, and the polynomial vanishes. This constructs all ten positive inputs freshly and proves both existential directions.

4.7 Exact cost, degree, and checks

The actual circuits have the following operation counts:

Weight	Multiplications	Additions	Subtractions	Total	Degree
V^2	46	18	14	78	630
bV	47	18	14	79	468

The relevant global degrees are

register	A	Δ	E	U	a	F_*	Q	G	V
deg	6	12	22	11	56	112	124	78	118

with $\deg b = 6$, $\deg M = 15$, and $\deg B = 2$. For $L = V^2$, $\deg Z = 253$ and QZ^2 uniquely supplies degree 630. For $L = bV$, $\deg Z = 141$ and $-QM^2GV^2$ uniquely supplies degree 468.

Let $q_0 = \beta J$, $k_0 = \eta + \zeta$, and $t_0 = T + \alpha$. The entire leading homogeneous components are

$$[P_{V^2}]_{630} = 2^{14} q_0^{382} w^{46} s^{112} i^{20} k_0^{64} t_0^4 (q_0 - F - \alpha)^2,$$
$$[P_{bV}]_{468} = -2^{10} q_0^{288} w^{34} s^{84} i^{14} k_0^{46} t_0^2.$$

They are nonzero at the actual fixed positive β. In particular the displayed formulas contain coefficients $2^{14}\beta^{382}$ and $-2^{10}\beta^{288}$ on the respective monomials

$$J^{382} w^{46} s^{112} i^{20} \eta^{64} T^4 F^2, \qquad J^{288} w^{34} s^{84} i^{14} \eta^{46} T^2.$$

These also illustrate the cost of the compact representation: the expanded coefficients can be much larger than the stored ports.

The independent standard-library checker proves 28 exact polynomial cut identities per circuit, covering all 78 or 79 actual registers. It verifies topology, backward liveness, every fixed coefficient byte, the complete leading forms (1,950 and 141 terms), and eight actual algebraic corruptions per circuit. Normal and optimized Python checks and byte-identical circuit rebuilds are supplied separately. These finite checks verify the artifacts and their algebra; the unbounded zero-existence equivalence is the ordinary proof above and its stated inherited interfaces.

5 What the verification proves

There are three distinct levels of evidence.

Ordinary unbounded proofs. The new Pell, valuation, factorial, CRT, and cap arguments establish the quantified mathematical implications. They compose with named inherited arithmetic and logical interfaces. A finite test is not used to infer a statement about all polynomial zeros.

Exact finite algebra. The supplied independent checkers reconstruct every coefficient of both quartics, verify every retained affine or monic elimination and its restoration, check the integral translation in the signed branch, and verify the compact circuits' topology, liveness, coefficient binding, operation counts, and complete leading homogeneous forms. The natural exporter independently checks 626 defining gates, 400 nonnegative affine restorations, and all 288 normalized rows before the 106 further contractions. Its two nontrivial positivity restorations are verified by exact rational sum-of-squares identities.

The natural reconstruction is replayed from the included relocated arithmetic engine and its literal numerical-calculus input. The signed replay starts from the included, pinned factored source of package 131; it verifies the new source changes, projection, and final translation. That starting boundary is deliberate and documented: it is not presented as a rerun of every historical compiler stage. The compact coefficients are reconstructed from their finite tile recipe, rather than accepted from hashes alone. Checks run with explicit exceptions, and both normal and optimized Python are exercised. Deliberately corrupted coefficients or circuit data are rejected, including a natural coefficient alteration whose claimed hash has been updated.

Logical/formal boundary. We inspected and retained the concrete refutation interfaces and the ordinary compiler proofs on which the new equations depend. We did not rebuild the full transitive Lean development, generate a Rocq proof, or produce an encoded object-language ZFC derivation of the new equations. The result is an ordinary mathematical construction formalizable in ZFC, with exact machine-checked finite algebra; it is not a newly completed end-to-end proof-assistant development. The inherited interface names do not turn the accompanying Python programs into a logical kernel.

5.1 Reproduction

From the extracted bundle root, using Python 3.10 or later:

```
python3 -B verify.py
python3 -B -O verify.py
```

The root runner records the checks for each component. Individual component READMEs state their exact replay boundaries and commands. The ordinary mathematical arguments remain readable independently of those programs. No search for a contradiction or a polynomial solution is performed.

The complete paper can be compiled from `report/report.tex` with pdfLaTeX twice. The final PDF has been rendered and visually inspected; the archive also includes the full editable TeX sources. File hashes and the immutable repository comparison commit are in the provenance files. The data encoding for sparse polynomials is `[coefficient,[coordinateindices]]`; repeated indices are repeated factors, and an empty index list denotes a constant. All coordinate indices start at zero. Thus the supplied files specify literal ordinary polynomials without factorials, variable exponents, floor functions, bounded-quantifier macros, or unspecified universal polynomials.

6 Conclusions and the remaining target

The strongest result here in the direct small-coefficient integer branch is the 314-variable quartic. The new polynomial comparison center is reusable wherever an indexed-Pell certificate already contains its auxiliary norm. The natural branch gives a separate useful improvement with a particularly short new factorial proof. The compact branch shows that a denominator introduced to make a proof convenient can sometimes be removed by changing the order in which arithmetic information is recovered.

None of these observations supplies a lower bound establishing the globally simplest consistency equation. The direct and compact constructions still make different trades between number of unknowns, degree, coefficient size, expanded support, and evaluation length. Further useful work should preserve all these measures, together with the domain and the exact calibration, when comparing a candidate. The included literal equations and proof boundaries make such a comparison reproducible.

References

[1] V. Reshetnikov, *ProveIt*, comparison commit `1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61`. https://github.com/VladimirReshetnikov/ProveIt.

[2] ProveIt package 129, *Semantic Anchors for Diophantine Consistency*, 24 October 2026, especially the natural 233/234 equations and ordinary arithmetic proof. https://github.com/VladimirReshetnikov/ProveIt/tree/1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61/Computability/DiophantineConsistency/con-zfc-diophantine-forms/129-semantic-pell-anchor.

[3] ProveIt package 131, *Shared Pell Caps and Coprime Product Equations for ZFC Consistency*, signed 324-variable endpoint and its retained calibration, same pinned commit.

[4] ProveIt package 132, *Signed Pell Caps and Free Scale Carriers*, odd-quotient and signed-cap arguments, same pinned commit.

[5] ProveIt package 133, *Staged Pell Certificates and a Product-Square Equation for Con(ZFC)*, compact 84/436 and 85/328 endpoints and complete coefficient tuple, same pinned commit.

[6] ProveIt package 128, *One Quadratic Constraint for Many Pell Certificates*, one-row cap theorem, same pinned commit.

[7] J. P. Jones, Universal Diophantine equation, *Journal of Symbolic Logic* **47** (1982), 549–571.

[8] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2 September 2026. https://arxiv.org/abs/2609.01594.

A The complete unrestricted-integer residual list

All 314 variables $z_0, \dots, z_{313}$ range independently over $\mathbb{Z}$. The equation is $\sum_{j=0}^{203} r_j^2 = 0$, using every row below. No semantic abbreviation remains in this finite definition.

$$
\begin{aligned}
r_0 &= -z_3^2 + z_4 \\
r_1 &= -1 - z_0 z_2 + z_9 \\
r_2 &= -1 + z_1 + z_2 \\
&\quad -z_2 z_7 - z_3 - z_7 \\
r_3 &= -2 + z_1 - z_8 z_9 \\
r_4 &= 1 + z_4 - z_5 z_{20} \\
&\quad +z_{18} \\
r_5 &= -1 + 2z_{10} - z_{10}^2 \\
&\quad -2z_{10} z_{11} + 2z_{11} - z_{11}^2 \\
&\quad +z_{21} \\
r_6 &= -5 - 30z_{10} - 36z_{10}^2 \\
&\quad -96z_{10} z_{11} - 24z_{10} z_{21} - 40z_{11} \\
&\quad -64z_{11}^2 - 32z_{11} z_{21} - 10z_{21} \\
&\quad -4z_{21}^2 + z_{22} \\
r_7 &= -2 + 2z_{10} - z_{10}^2 \\
&\quad -4z_{10} z_{22} + 4z_{22} - 4z_{22}^2 \\
&\quad +z_{23} \\
r_8 &= 1 + 6z_{10} - 36z_{10}^2 \\
&\quad -192z_{10} z_{22} - 24z_{10} z_{23} + 16z_{22} \\
&\quad -256z_{22}^2 - 64z_{22} z_{23} + 2z_{23} \\
&\quad -4z_{23}^2 + z_{24} \\
r_9 &= -2 - z_{11}^2 - 2z_{11} z_{12} \\
&\quad -z_{12}^2 + z_{25} \\
r_{10} &= -9 + 7z_{10} - z_{10}^2 \\
&\quad -4z_{10} z_{11} - 4z_{10} z_{25} + 12z_{11} \\
&\quad -4z_{11}^2 - 8z_{11} z_{25} + 12z_{25} \\
&\quad -4z_{25}^2 + z_{26} \\
r_{11} &= -1 + 2z_{10} - z_{10}^2 \\
&\quad -2z_{10} z_{12} + 2z_{12} - z_{12}^2 \\
&\quad +z_{27} \\
r_{12} &= -6z_{10} - 144z_{10}^2 - 192z_{10} z_{11} \\
&\quad -192z_{10} z_{12} - 48z_{10} z_{21} - 48z_{10} z_{27} \\
&\quad -8z_{11} - 64z_{11}^2 - 128z_{11} z_{12} \\
&\quad -32z_{11} z_{21} - 32z_{11} z_{27} - 64z_{12}^2 \\
&\quad -32z_{12} z_{21} - 32z_{12} z_{27} - 2z_{21} \\
&\quad -4z_{21}^2 - 8z_{21} z_{27} - 4z_{27}^2 \\
&\quad +z_{28}
\end{aligned}
$$

$$
\begin{aligned}
r_{13} &= -4z_{26}^2 - 8z_{26}z_{28} - 4z_{28}^2 \\
&\quad +z_{29} \\
r_{14} &= -6 + 16z_{12} - 16z_{12}^2 \\
&\quad -16z_{12}z_{25} - 16z_{12}z_{27} + 8z_{25} \\
&\quad -4z_{25}^2 - 8z_{25}z_{27} + 8z_{27} \\
&\quad -4z_{27}^2 + z_{30} \\
r_{15} &= 1 + 6z_{10} - 36z_{10}^2 \\
&\quad -96z_{10}z_{11} - 48z_{10}z_{12} - 24z_{10}z_{21} \\
&\quad -48z_{10}z_{27} - 24z_{10}z_{30} + 8z_{11} \\
&\quad -64z_{11}^2 - 64z_{11}z_{12} - 32z_{11}z_{21} \\
&\quad -64z_{11}z_{27} - 32z_{11}z_{30} + 8z_{12} \\
&\quad -16z_{12}^2 - 16z_{12}z_{21} - 32z_{12}z_{27} \\
&\quad -16z_{12}z_{30} + 2z_{21} - 4z_{21}^2 \\
&\quad -16z_{21}z_{27} - 8z_{21}z_{30} + 8z_{27} \\
&\quad -16z_{27}^2 - 16z_{27}z_{30} + 4z_{30} \\
&\quad -4z_{30}^2 + z_{31} \\
r_{16} &= -10 - 5z_{10} - z_{10}^2 \\
&\quad -12z_{10}z_{11} - 16z_{10}z_{12} - 4z_{10}z_{25} \\
&\quad -36z_{11} - 36z_{11}^2 - 96z_{11}z_{12} \\
&\quad -24z_{11}z_{25} - 48z_{12} - 64z_{12}^2 \\
&\quad -32z_{12}z_{25} - 12z_{25} - 4z_{25}^2 \\
&\quad +z_{32} \\
r_{17} &= -2 - 24z_{10} - 36z_{10}^2 \\
&\quad -12z_{10}z_{11} - 96z_{10}z_{12} - 24z_{10}z_{27} \\
&\quad -3z_{11} - z_{11}^2 - 16z_{11}z_{12} \\
&\quad -4z_{11}z_{27} - 32z_{12} - 64z_{12}^2 \\
&\quad -32z_{12}z_{27} - 8z_{27} - 4z_{27}^2 \\
&\quad +z_{33} \\
r_{18} &= 2 - 4z_{32}^2 - 8z_{32}z_{33} \\
&\quad -4z_{33}^2 + z_{34} \\
r_{19} &= -12 + 16z_{10} - 4z_{10}^2 \\
&\quad -8z_{10}z_{12} - 8z_{10}z_{21} - 8z_{10}z_{25} \\
&\quad +16z_{12} - 4z_{12}^2 - 8z_{12}z_{21} \\
&\quad -8z_{12}z_{25} + 16z_{21} - 4z_{21}^2 \\
&\quad -8z_{21}z_{25} + 16z_{25} - 4z_{25}^2 \\
&\quad +z_{35} \\
r_{20} &= -3 - 24z_{10} - 64z_{10}^2 \\
&\quad -160z_{10}z_{12} - 64z_{10}z_{25} - 32z_{10}z_{27} \\
&\quad -32z_{10}z_{35} - 26z_{12} - 100z_{12}^2 \\
&\quad -80z_{12}z_{25} - 40z_{12}z_{27} - 40z_{12}z_{35} \\
&\quad -8z_{25} - 16z_{25}^2 - 16z_{25}z_{27} \\
&\quad -16z_{25}z_{35} - 6z_{27} - 4z_{27}^2 \\
&\quad -8z_{27}z_{35} - 4z_{35} - 4z_{35}^2 \\
&\quad +z_{36} \\
r_{21} &= -1 - 18z_{10} - 36z_{10}^2 \\
&\quad -96z_{10}z_{11} - 24z_{10}z_{21} - 24z_{11} \\
&\quad -64z_{11}^2 - 32z_{11}z_{21} - 6z_{21} \\
&\quad -4z_{21}^2 + z_{37} \\
r_{22} &= -2 + 2z_{10} - z_{10}^2 \\
&\quad -4z_{10}z_{37} + 4z_{37} - 4z_{37}^2 \\
&\quad +z_{38} \\
r_{23} &= 1 + 6z_{10} - 36z_{10}^2 \\
&\quad -192z_{10}z_{37} - 24z_{10}z_{38} + 16z_{37} \\
&\quad -256z_{37}^2 - 64z_{37}z_{38} + 2z_{38} \\
&\quad -4z_{38}^2 + z_{39}
\end{aligned}
$$

$$
\begin{aligned}
r_{24} = &-z_{11}^2 - 4z_{11}z_{39} - 4z_{39}^2 \\
&+z_{40} \\
r_{25} = &-6 - 30z_{10} - 64z_{10}^2 \\
&-128z_{10}z_{11} - 64z_{10}z_{21} - 32z_{11} \\
&-64z_{11}^2 - 64z_{11}z_{21} - 14z_{21} \\
&-16z_{21}^2 + z_{41} \\
r_{26} = &-4 + 8z_{11} - 64z_{11}^2 \\
&-96z_{11}z_{12} - 32z_{11}z_{25} - 32z_{11}z_{41} \\
&+6z_{12} - 36z_{12}^2 - 24z_{12}z_{25} \\
&-24z_{12}z_{41} + 2z_{25} - 4z_{25}^2 \\
&-8z_{25}z_{41} + 4z_{41} - 4z_{41}^2 \\
&+z_{42} \\
r_{27} = &\,8z_{10} - 64z_{10}^2 - 96z_{10}z_{12} \\
&-32z_{10}z_{27} - 32z_{10}z_{42} + 6z_{12} \\
&-36z_{12}^2 - 24z_{12}z_{27} - 24z_{12}z_{42} \\
&+2z_{27} - 4z_{27}^2 - 8z_{27}z_{42} \\
&+4z_{42} - 4z_{42}^2 + z_{43} \\
r_{28} = &-z_{12}^2 - 2z_{12}z_{13} - z_{13}^2 \\
&+z_{44} \\
r_{29} = &\,2z_{10} - z_{10}^2 - 4z_{10}z_{12} \\
&-4z_{10}z_{44} + 4z_{12} - 4z_{12}^2 \\
&-8z_{12}z_{44} + 4z_{44} - 4z_{44}^2 \\
&+z_{45} \\
r_{30} = &-35 + 24z_{10} - 4z_{10}^2 \\
&-8z_{10}z_{11} - 8z_{10}z_{25} - 8z_{10}z_{27} \\
&+24z_{11} - 4z_{11}^2 - 8z_{11}z_{25} \\
&-8z_{11}z_{27} + 24z_{25} - 4z_{25}^2 \\
&-8z_{25}z_{27} + 24z_{27} - 4z_{27}^2 \\
&+z_{46} \\
r_{31} = &-16z_{10}^2 - 32z_{10}z_{12} - 32z_{10}z_{27} \\
&-32z_{10}z_{44} - 16z_{10}z_{45} - 16z_{10}z_{46} \\
&-4z_{12} - 16z_{12}^2 - 32z_{12}z_{27} \\
&-32z_{12}z_{44} - 16z_{12}z_{45} - 16z_{12}z_{46} \\
&-16z_{27}^2 - 32z_{27}z_{44} - 16z_{27}z_{45} \\
&-16z_{27}z_{46} - 4z_{44} - 16z_{44}^2 \\
&-16z_{44}z_{45} - 16z_{44}z_{46} - 2z_{45} \\
&-4z_{45}^2 - 8z_{45}z_{46} - 4z_{46}^2 \\
&+z_{47} \\
r_{32} = &\,z_{12} - 25z_{12}^2 - 40z_{12}z_{27} \\
&-20z_{12}z_{30} - 16z_{27}^2 - 16z_{27}z_{30} \\
&-4z_{30}^2 + z_{48} \\
r_{33} = &-1 + 8z_{11} - 4z_{11}^2 \\
&-8z_{11}z_{25} + 8z_{25} - 4z_{25}^2 \\
&+z_{49} \\
r_{34} = &-1 - 16z_{11}^2 - 8z_{11}z_{12} \\
&-32z_{11}z_{25} - 16z_{11}z_{49} - z_{12}^2 \\
&-8z_{12}z_{25} - 4z_{12}z_{49} - 16z_{25}^2 \\
&-16z_{25}z_{49} - 4z_{49}^2 + z_{50} \\
r_{35} = &\,1 + z_{11} - z_{11}^2 \\
&-4z_{11}z_{12} - 4z_{11}z_{50} - 4z_{12}^2 \\
&-8z_{12}z_{50} - 4z_{50}^2 + z_{51} \\
r_{36} = &-1 - 4z_{51} - 4z_{51}^2 \\
&+z_{52} \\
r_{37} = &-1 - z_{12}^2 - 8z_{12}z_{51} \\
&-4z_{12}z_{52} - 16z_{51}^2 - 16z_{51}z_{52}
\end{aligned}
$$

$$
\begin{aligned}
& -4z_{52}^2 + z_{53} \\
r_{38} = {} & 2 + z_{11} - z_{11}^2 \\
& -4z_{11}z_{12} - 4z_{11}z_{53} - 4z_{12}^2 \\
& -8z_{12}z_{53} - 4z_{53}^2 + z_{54} \\
r_{39} = {} & -4z_{48}^2 - 8z_{48}z_{54} - 4z_{54}^2 \\
& +z_{55} \\
r_{40} = {} & -9 + 7z_{10} - z_{10}^2 \\
& -8z_{10}z_{48} - 4z_{10}z_{55} + 24z_{48} \\
& -16z_{48}^2 - 16z_{48}z_{55} + 12z_{55} \\
& -4z_{55}^2 + z_{56} \\
r_{41} = {} & -1 - 4z_{56} - 4z_{56}^2 \\
& +z_{57} \\
r_{42} = {} & -z_{12}^2 - 8z_{12}z_{56} - 4z_{12}z_{57} \\
& -16z_{56}^2 - 16z_{56}z_{57} - 4z_{57}^2 \\
& +z_{58} \\
r_{43} = {} & -z_{10} - z_{10}^2 - 4z_{10}z_{12} \\
& -4z_{10}z_{58} - 4z_{12} - 4z_{12}^2 \\
& -8z_{12}z_{58} - 4z_{58} - 4z_{58}^2 \\
& +z_{59} \\
r_{44} = {} & -10 - 12z_{59} - 4z_{59}^2 \\
& +z_{60} \\
r_{45} = {} & 1 + 4z_{11} - 4z_{11}^2 \\
& -8z_{11}z_{25} - 16z_{11}z_{59} - 8z_{11}z_{60} \\
& +4z_{25} - 4z_{25}^2 - 16z_{25}z_{59} \\
& -8z_{25}z_{60} + 8z_{59} - 16z_{59}^2 \\
& -16z_{59}z_{60} + 4z_{60} - 4z_{60}^2 \\
& +z_{61} \\
r_{46} = {} & z_{11} - 25z_{11}^2 - 40z_{11}z_{25} \\
& -20z_{11}z_{61} - 16z_{25}^2 - 16z_{25}z_{61} \\
& -4z_{61}^2 + z_{62} \\
r_{47} = {} & -10 - 12z_{54} - 4z_{54}^2 \\
& +z_{63} \\
r_{48} = {} & 1 - 16z_{54}^2 - 16z_{54}z_{62} \\
& -16z_{54}z_{63} - 4z_{62}^2 - 8z_{62}z_{63} \\
& -4z_{63}^2 + z_{64} \\
r_{49} = {} & z_{12} - z_{12}^2 - 8z_{12}z_{62} \\
& -4z_{12}z_{64} - 16z_{62}^2 - 16z_{62}z_{64} \\
& -4z_{64}^2 + z_{65} \\
r_{50} = {} & -4z_{10} - 4z_{10}^2 - 4z_{10}z_{12} \\
& -8z_{10}z_{21} - z_{12} - z_{12}^2 \\
& -4z_{12}z_{21} - 4z_{21} - 4z_{21}^2 \\
& +z_{66} \\
r_{51} = {} & 1 + 4z_{10} - 4z_{10}^2 \\
& -8z_{10}z_{21} - 8z_{10}z_{66} + 4z_{21} \\
& -4z_{21}^2 - 8z_{21}z_{66} + 4z_{66} \\
& -4z_{66}^2 + z_{67} \\
r_{52} = {} & -2 - 4z_{10} - 16z_{10}^2 \\
& -32z_{10}z_{21} - 16z_{10}z_{65} - 16z_{10}z_{67} \\
& -4z_{21} - 16z_{21}^2 - 16z_{21}z_{65} \\
& -16z_{21}z_{67} - 4z_{65}^2 - 8z_{65}z_{67} \\
& -2z_{67} - 4z_{67}^2 + z_{68} \\
r_{53} = {} & -2 - 4z_{68} - 4z_{68}^2 \\
& +z_{69} \\
r_{54} = {} & -z_3z_{15} + z_{70}
\end{aligned}
$$

$$
\begin{aligned}
r_{55} &= z_{14} - z_{15} - 2z_{24}z_{70} \\
&\quad -z_{71} \\
r_{56} &= z_{14} - z_{15} - 4z_{26}z_{70} \\
&\quad -2z_{29}z_{70} - z_{72} \\
r_{57} &= z_{14} - z_{15} - 2z_{31}z_{70} \\
&\quad -6z_{70} - z_{73} \\
r_{58} &= z_{14} - z_{15} - 4z_{32}z_{70} \\
&\quad -2z_{34}z_{70} - z_{74} \\
r_{59} &= z_{14} - z_{15} - 2z_{36}z_{70} \\
&\quad +6z_{70} - z_{75} \\
r_{60} &= z_{14} - z_{15} - 4z_{39}z_{70} \\
&\quad -2z_{40}z_{70} - z_{76} \\
r_{61} &= z_{14} - z_{15} - 2z_{43}z_{70} \\
&\quad -z_{77} \\
r_{62} &= z_{14} - z_{15} - 2z_{47}z_{70} \\
&\quad -4z_{70} - z_{78} \\
r_{63} &= -z_{10}z_{11} + z_{10}z_{12} + z_{11} \\
&\quad +z_{11}z_{12} - z_{12} - z_{12}^2 \\
&\quad +z_{79} \\
r_{64} &= 1 + 2z_{13} + z_{13}^2 \\
&\quad -z_{79}^2 + z_{80} \\
r_{65} &= -z_{80}^2 + z_{81} \\
r_{66} &= -4z_{68}z_{70} - 2z_{69}z_{70} + 2z_{70} \\
&\quad +z_{82} \\
r_{67} &= -z_{14}^2 + 2z_{14}z_{15} + 2z_{14}z_{82} \\
&\quad -z_{15}^2 - 2z_{15}z_{82} - z_{82}^2 \\
&\quad +z_{83} \\
r_{68} &= -1 - z_3z_{11} + z_{84} \\
r_{69} &= -z_{15}z_{84} + z_{86}z_{87} + z_{88} \\
r_{70} &= -1 + z_3 - z_3z_{10} \\
&\quad +z_{85} \\
r_{71} &= -1 + 2z_3 - 2z_3z_{10} \\
&\quad -2z_3z_{21} + z_{86} \\
r_{72} &= -z_{14}z_{85} + z_{87} \\
r_{73} &= -1 - 6z_3z_{10} - 8z_3z_{11} \\
&\quad -2z_3z_{21} + z_{89} \\
r_{74} &= z_{14}z_{84} - z_{15}z_{89} + z_{90} \\
r_{75} &= -z_{14}z_{71} + z_{15}z_{71} + z_{91} \\
r_{76} &= -z_{72}z_{91} + z_{92} \\
r_{77} &= -z_{73}z_{92} + z_{93} \\
r_{78} &= -z_{74}z_{93} + z_{94} \\
r_{79} &= -z_{75}z_{94} + z_{95} \\
r_{80} &= -z_{76}z_{95} + z_{96} \\
r_{81} &= -z_{77}z_{96} + z_{97} \\
r_{82} &= -z_{78}z_{97} + z_{98} \\
r_{83} &= -z_{88}z_{98} + z_{99} \\
r_{84} &= -z_{90}z_{99} + z_{100} \\
r_{85} &= -z_{100}^2 + z_{101} \\
r_{86} &= -1 - z_2 - z_2z_{18} \\
&\quad +z_{102}
\end{aligned}
$$

$$
\begin{aligned}
r_{87} &= -1 - z_{16}z_{102} + z_{103} \\
r_{88} &= -2z_1^2 + 2z_1z_{14} + 2z_1z_{15} \\
&\quad +2z_1z_{103} + 2z_1z_{104} - z_{14}^2 \\
&\quad -2z_{14}z_{103} - z_{15}^2 - 2z_{15}z_{104} \\
&\quad +z_{19}z_{106} - z_{81}z_{101} - z_{83}z_{101} \\
&\quad -z_{103}^2 - z_{104}^2 \\
r_{89} &= -1 - z_2z_{17} - z_{17}z_{102} \\
&\quad +z_{104} \\
r_{90} &= -9z_{10} - 9z_{11} - 4z_{21} \\
&\quad -2z_{24} - 4z_{26} - 2z_{29} \\
&\quad -2z_{31} - 4z_{32} - 2z_{34} \\
&\quad -2z_{36} - 4z_{39} - 2z_{40} \\
&\quad -2z_{43} - 2z_{47} - 4z_{68} \\
&\quad -2z_{69} + z_{105} \\
r_{91} &= -z_{107}z_{110} + z_{110} + z_{112} \\
r_{92} &= -z_5z_{109} + z_{108} - z_{109}z_{112} \\
&\quad -z_{111} \\
r_{93} &= -z_{19}z_{114} + z_{117} \\
r_{94} &= z_{113} - z_{115}z_{117} - z_{116} \\
r_{95} &= -z_{117}z_{119} + z_{118} - z_{120} \\
r_{96} &= -z_{117}z_{122} + z_{121} - z_{123} \\
r_{97} &= -z_{117}z_{125} + z_{124} - z_{126} \\
r_{98} &= -z_{19}z_{128} + z_{131} \\
r_{99} &= z_{127} - z_{129}z_{131} - z_{130} \\
r_{100} &= -z_{131}z_{133} + z_{132} - z_{134} \\
r_{101} &= -z_{19}z_{136} + z_{139} \\
r_{102} &= z_{135} - z_{137}z_{139} - z_{138} \\
r_{103} &= -z_{139}z_{141} + z_{140} - z_{142} \\
r_{104} &= -z_{19}z_{109} + z_{146} \\
r_{105} &= z_{143} - z_{144}z_{146} - z_{145} \\
r_{106} &= -z_{117}z_{148} + z_{147} - z_{149} \\
r_{107} &= z_{150} - 2z_{150}z_{152} + z_{153} \\
r_{108} &= -z_{150}z_{153} - z_{153} + z_{154} \\
r_{109} &= 2z_{151} + z_{151}^2 + z_{153}^2 \\
&\quad -z_{154}^2 \\
r_{110} &= 1 + z_{151} - z_{151}z_{156} \\
&\quad -z_{156} + z_{157} \\
r_{111} &= z_{155}^2 - 2z_{156} + z_{156}^2 \\
&\quad -z_{157}^2 \\
r_{112} &= z_4^2 - 2z_4z_{151} - 2z_{151} \\
&\quad +z_{158} \\
r_{113} &= 1 + z_4 - z_4z_{156} \\
&\quad +z_6 - z_{155} - z_{156} \\
&\quad +z_{157} + z_{158}z_{159} \\
r_{114} &= -z_{151}z_{161} - z_{161} + z_{162} \\
r_{115} &= -1 + z_{160}^2 + z_{161}^2 \\
&\quad -z_{162}^2 \\
r_{116} &= 1 - 2z_5 + z_5^2 \\
&\quad -2z_5z_{151} + z_{163} \\
r_{117} &= -z_5z_{161} + z_{19} - z_{160} \\
&\quad +z_{162} + z_{163}z_{164}
\end{aligned}
$$

$$
\begin{aligned}
r_{118} &= -1 + 2z_{151}z_{224} - z_{165} \\
&\quad +2z_{224} - z_{224}^2 \\
r_{119} &= -1 + z_{107} - z_{160} \\
&\quad -z_{161}z_{224} + z_{162} + z_{165}z_{166} \\
r_{120} &= 1 + z_{151} - z_{151}z_{168} \\
&\quad -z_{168} + z_{169} \\
r_{121} &= z_{167}^2 - 2z_{168} + z_{168}^2 \\
&\quad -z_{169}^2 \\
r_{122} &= 1 - 2z_{107} + z_{107}^2 \\
&\quad -2z_{107}z_{151} + z_{170} \\
r_{123} &= z_{107} - z_{107}z_{168} + z_{108} \\
&\quad -z_{167} + z_{169} + z_{170}z_{171} \\
r_{124} &= 4 - 4z_{107} + z_{107}^2 \\
&\quad -2z_{107}z_{151} + 2z_{151} + z_{172} \\
r_{125} &= -1 + z_{107} - z_{107}z_{156} \\
&\quad +z_{109} - z_{155} + z_{156} \\
&\quad +z_{157} + z_{172}z_{173} \\
r_{126} &= 1 + z_{151} - z_{151}z_{175} \\
&\quad -z_{175} + z_{176} \\
r_{127} &= z_{174}^2 - 2z_{175} + z_{175}^2 \\
&\quad -z_{176}^2 \\
r_{128} &= z_{107} - z_{107}z_{175} + z_{113} \\
&\quad +z_{170}z_{177} - z_{174} + z_{176} \\
r_{129} &= -2z_{151} - z_{151}^2 + z_{178} \\
r_{130} &= -z_{107}z_{155} - z_{107}z_{157} + z_{114} \\
&\quad +z_{151}z_{157} + z_{155} - z_{156}z_{178} \\
&\quad +2z_{157} + z_{172}z_{179} + z_{178} \\
r_{131} &= -z_{151}z_{181} - z_{181} + z_{182} \\
r_{132} &= -1 + z_{180}^2 + z_{181}^2 \\
&\quad -z_{182}^2 \\
r_{133} &= -z_{107}z_{181} + z_{118} + z_{170}z_{183} \\
&\quad -z_{180} + z_{182} \\
r_{134} &= -z_{151}z_{185} - z_{185} + z_{186} \\
r_{135} &= -1 + z_{184}^2 + z_{185}^2 \\
&\quad -z_{186}^2 \\
r_{136} &= -z_{107}z_{185} + z_{121} + z_{170}z_{187} \\
&\quad -z_{184} + z_{186} \\
r_{137} &= -z_{151}z_{189} - z_{189} + z_{190} \\
r_{138} &= -1 + z_{188}^2 + z_{189}^2 \\
&\quad -z_{190}^2 \\
r_{139} &= -z_{107}z_{189} + z_{124} + z_{170}z_{191} \\
&\quad -z_{188} + z_{190} \\
r_{140} &= -z_{151}z_{193} - z_{193} + z_{194} \\
r_{141} &= -1 + z_{192}^2 + z_{193}^2 \\
&\quad -z_{194}^2 \\
r_{142} &= -z_{107}z_{193} + z_{127} + z_{170}z_{195} \\
&\quad -z_{192} + z_{194} \\
r_{143} &= -1 - z_{151} - z_{151}z_{197} \\
&\quad -z_{197} + z_{198} \\
r_{144} &= z_{196}^2 + 2z_{197} + z_{197}^2 \\
&\quad -z_{198}^2 \\
r_{145} &= 1 - z_{107} - z_{107}z_{197}
\end{aligned}
$$

$$
\begin{aligned}
&+z_{128} + z_{172}z_{199} - z_{196} \\
&+z_{197} + z_{198} \\
r_{146} &= -z_{151}z_{201} - z_{201} + z_{202} \\
r_{147} &= -1 + z_{200}^2 + z_{201}^2 \\
&-z_{202}^2 \\
r_{148} &= -z_{107}z_{201} + z_{132} + z_{170}z_{203} \\
&-z_{200} + z_{202} \\
r_{149} &= -z_{151}z_{205} - z_{205} + z_{206} \\
r_{150} &= -1 + z_{204}^2 + z_{205}^2 \\
&-z_{206}^2 \\
r_{151} &= -z_{107}z_{205} + z_{135} + z_{170}z_{207} \\
&-z_{204} + z_{206} \\
r_{152} &= -z_{151}z_{209} - z_{209} + z_{210} \\
r_{153} &= -1 + z_{208}^2 + z_{209}^2 \\
&-z_{210}^2 \\
r_{154} &= -z_{107}z_{209} + z_{136} + z_{172}z_{211} \\
&-z_{208} + z_{209} + z_{210} \\
r_{155} &= -z_{151}z_{213} - z_{213} + z_{214} \\
r_{156} &= -1 + z_{212}^2 + z_{213}^2 \\
&-z_{214}^2 \\
r_{157} &= -z_{107}z_{213} + z_{140} + z_{170}z_{215} \\
&-z_{212} + z_{214} \\
r_{158} &= -z_{151}z_{217} - z_{217} + z_{218} \\
r_{159} &= -1 + z_{216}^2 + z_{217}^2 \\
&-z_{218}^2 \\
r_{160} &= -z_{107}z_{217} + z_{143} + z_{170}z_{219} \\
&-z_{216} + z_{218} \\
r_{161} &= -z_{151}z_{221} - z_{221} + z_{222} \\
r_{162} &= -1 + z_{220}^2 + z_{221}^2 \\
&-z_{222}^2 \\
r_{163} &= -z_{107}z_{221} + z_{147} + z_{170}z_{223} \\
&-z_{220} + z_{222} \\
r_{164} &= -16 + 8z_0 - 14z_0^2 \\
&-16z_0z_1 - 16z_0z_2 - 16z_0z_3 \\
&-4z_0z_4 - 10z_1^2 - 16z_1z_2 \\
&-16z_1z_3 - 4z_2 - 10z_2^2 \\
&-16z_2z_3 - 8z_3^2 - 4z_4^2 \\
&-2z_5^2 - 2z_6^2 + 4z_{10} \\
&-2z_{10}^2 - 2z_{11}^2 - 2z_{12}^2 \\
&-2z_{13}^2 - 2z_{14}^2 - 2z_{15}^2 \\
&-2z_{16}^2 - 2z_{17}^2 - 2z_{18}^2 \\
&-2z_{19}^2 - 2z_{105}^2 + 4z_{107} \\
&-4z_{107}^2 - 2z_{108}^2 - 2z_{109}^2 \\
&-2z_{113}^2 - 2z_{114}^2 - 2z_{118}^2 \\
&-2z_{121}^2 - 2z_{124}^2 - 2z_{127}^2 \\
&-2z_{128}^2 - 2z_{132}^2 - 2z_{135}^2 \\
&-2z_{136}^2 - 2z_{140}^2 - 2z_{143}^2 \\
&-2z_{147}^2 + z_{150} - 2z_{224}^2 \\
r_{165} &= -z_5z_{10} - z_5z_{11} - z_5z_{12} \\
&-z_5z_{13} - z_5z_{14} - z_5z_{15} \\
&-z_5z_{16} - z_5z_{17} - z_5z_{18} \\
&+z_{224} \\
r_{166} &= -17 + z_0 - z_{225}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{167} &= z_2 + z_6 - z_6^2 \\
r_{168} &= -z_{151}z_{227} - z_{227} + z_{234} \\
r_{169} &= -1 + z_{226}^2 + z_{227}^2 \\
&\quad -z_{234}^2 \\
r_{170} &= -z_{229} - z_{229}z_{230} + z_{235} \\
r_{171} &= -1 + z_{228}^2 + z_{229}^2 \\
&\quad -z_{235}^2 \\
r_{172} &= -z_{221}^2 + z_{236} \\
r_{173} &= z_{227} - 2z_{231}z_{236} - z_{236} \\
r_{174} &= -z_{220} - z_{226}z_{232} + z_{228} \\
r_{175} &= -z_{105} - 4z_{221}z_{233} + z_{229} \\
r_{176} &= 6z_0^2 + 8z_0z_1 + 8z_0z_2 \\
&\quad +8z_0z_3 + 2z_0z_4 - 2z_0z_{156} \\
&\quad -4z_0z_{161} - 2z_0z_{168} + 5z_1^2 \\
&\quad +8z_1z_2 + 8z_1z_3 - 4z_1z_{161} \\
&\quad -2z_1z_{209} + 5z_2^2 + 8z_2z_3 \\
&\quad -4z_2z_{161} - 2z_2z_{197} + 4z_3^2 \\
&\quad -4z_3z_{161} + z_4^2 - 2z_4z_{168} \\
&\quad +z_{10}^2 - 2z_{10}z_{175} + z_{11}^2 \\
&\quad -2z_{11}z_{181} + z_{12}^2 - 2z_{12}z_{185} \\
&\quad +z_{13}^2 - 2z_{13}z_{189} + z_{14}^2 \\
&\quad -2z_{14}z_{193} + z_{15}^2 - 2z_{15}z_{201} \\
&\quad +z_{16}^2 - 2z_{16}z_{205} + z_{17}^2 \\
&\quad -2z_{17}z_{213} + z_{18}^2 - 2z_{18}z_{217} \\
&\quad +z_{105}^2 - 2z_{105}z_{221} - z_{151}z_{237} \\
&\quad +z_{156}^2 + z_{161}^2 + z_{168}^2 \\
&\quad +z_{175}^2 + z_{181}^2 + z_{185}^2 \\
&\quad +z_{189}^2 + z_{193}^2 + z_{197}^2 \\
&\quad +z_{201}^2 + z_{205}^2 + z_{209}^2 \\
&\quad +z_{213}^2 + z_{217}^2 + z_{221}^2 \\
r_{177} &= -1 + z_1 - z_{238}^2 \\
&\quad -z_{239}^2 - z_{240} - z_{240}^2 \\
r_{178} &= -1 - z_3z_5 + z_5 \\
&\quad +z_6 - z_{241}^2 - z_{242}^2 \\
&\quad -z_{243} - z_{243}^2 \\
r_{179} &= z_3z_5 - z_6 - z_{244}^2 \\
&\quad -z_{245}^2 - z_{246} - z_{246}^2 \\
r_{180} &= -2 + z_{10} - z_{247}^2 \\
&\quad -z_{248}^2 - z_{249} - z_{249}^2 \\
r_{181} &= -1 + z_{11} - z_{250}^2 \\
&\quad -z_{251}^2 - z_{252} - z_{252}^2 \\
r_{182} &= -1 + z_{12} - z_{253}^2 \\
&\quad -z_{254}^2 - z_{255} - z_{255}^2 \\
r_{183} &= -1 + z_{13} - z_{256}^2 \\
&\quad -z_{257}^2 - z_{258} - z_{258}^2 \\
r_{184} &= -1 + z_{14} - z_{259}^2 \\
&\quad -z_{260}^2 - z_{261} - z_{261}^2 \\
r_{185} &= -1 + z_{15} - z_{262}^2 \\
&\quad -z_{263}^2 - z_{264} - z_{264}^2 \\
r_{186} &= -1 + z_{16} - z_{265}^2 \\
&\quad -z_{266}^2 - z_{267} - z_{267}^2 \\
r_{187} &= -1 + z_{17} - z_{268}^2 \\
&\quad -z_{269}^2 - z_{270} - z_{270}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{188} &= -1 + z_{18} - z_{271}^2 \\
&\quad - z_{272}^2 - z_{273} - z_{273}^2 \\
r_{189} &= -2 - z_5 + z_{107} \\
&\quad - z_{274}^2 - z_{275}^2 - z_{276} \\
&\quad - z_{276}^2 \\
r_{190} &= z_{109} z_{111} - z_{111} - z_{111}^2 \\
&\quad - z_{277}^2 - z_{278}^2 - z_{279} \\
&\quad - z_{279}^2 \\
r_{191} &= z_{114} z_{116} - z_{116} - z_{116}^2 \\
&\quad - z_{280}^2 - z_{281}^2 - z_{282} \\
&\quad - z_{282}^2 \\
r_{192} &= z_{114} z_{120} - z_{120} - z_{120}^2 \\
&\quad - z_{283}^2 - z_{284}^2 - z_{285} \\
&\quad - z_{285}^2 \\
r_{193} &= z_{114} z_{123} - z_{123} - z_{123}^2 \\
&\quad - z_{286}^2 - z_{287}^2 - z_{288} \\
&\quad - z_{288}^2 \\
r_{194} &= z_{114} z_{126} - z_{126} - z_{126}^2 \\
&\quad - z_{289}^2 - z_{290}^2 - z_{291} \\
&\quad - z_{291}^2 \\
r_{195} &= z_{128} z_{130} - z_{130} - z_{130}^2 \\
&\quad - z_{292}^2 - z_{293}^2 - z_{294} \\
&\quad - z_{294}^2 \\
r_{196} &= z_{128} z_{134} - z_{134} - z_{134}^2 \\
&\quad - z_{295}^2 - z_{296}^2 - z_{297} \\
&\quad - z_{297}^2 \\
r_{197} &= z_{136} z_{138} - z_{138} - z_{138}^2 \\
&\quad - z_{298}^2 - z_{299}^2 - z_{300} \\
&\quad - z_{300}^2 \\
r_{198} &= z_{136} z_{142} - z_{142} - z_{142}^2 \\
&\quad - z_{301}^2 - z_{302}^2 - z_{303} \\
&\quad - z_{303}^2 \\
r_{199} &= z_{109} z_{145} - z_{145} - z_{145}^2 \\
&\quad - z_{304}^2 - z_{305}^2 - z_{306} \\
&\quad - z_{306}^2 \\
r_{200} &= z_{114} z_{149} - z_{149} - z_{149}^2 \\
&\quad - z_{307}^2 - z_{308}^2 - z_{309} \\
&\quad - z_{309}^2 \\
r_{201} &= -5 - z_{150}^2 + 2 z_{156} \\
&\quad - z_{156}^2 - z_{161}^2 + 2 z_{168} \\
&\quad - z_{168}^2 + 2 z_{175} - z_{175}^2 \\
&\quad - z_{181}^2 - z_{185}^2 - z_{189}^2 \\
&\quad - z_{193}^2 - 2 z_{197} - z_{197}^2 \\
&\quad - z_{201}^2 - z_{205}^2 - z_{209}^2 \\
&\quad - z_{213}^2 - z_{217}^2 + z_{221}^2 \\
&\quad - z_{310}^2 - z_{311}^2 - z_{312} \\
&\quad - z_{312}^2 \\
r_{202} &= 1 - z_{226}^2 + z_{313} \\
r_{203} &= z_{151} z_{313} + z_{230}
\end{aligned}
$$

B The complete natural-number residual list

All 233 variables $x_0, \dots, x_{232}$ range independently over $\mathbb{N}$. The equation is $\sum_{j=0}^{181} s_j^2 = 0$, using every row below. The variable-name map in the artifact is explanatory only; the indexed list is already a complete polynomial specification.

$$
\begin{aligned}
s_0 &= -3 - 4x_2 - x_2^2 \\
&\quad +x_3 \\
s_1 &= 1 + x_4 - x_5 \\
&\quad -x_6 \\
s_2 &= -39 - 3x_0 - x_0x_2 \\
&\quad -x_0x_8 - 13x_2 - 13x_8 \\
&\quad +x_{10} \\
s_3 &= -3 + x_1 - x_2 \\
&\quad -x_2x_7 - 4x_7 - x_7x_8 \\
s_4 &= -1 + x_1 - x_9 \\
&\quad -x_9x_{10} \\
s_5 &= 1 - x_3 - x_4x_{21} \\
&\quad +x_{19} - 2x_{21} \\
s_6 &= -6 - 5x_{11} - x_{11}^2 \\
&\quad -2x_{11}x_{12} - 5x_{12} - x_{12}^2 \\
&\quad +x_{22} \\
s_7 &= -72 - 68x_{12} - 16x_{12}^2 \\
&\quad -16x_{12}x_{22} - 34x_{22} - 4x_{22}^2 \\
&\quad +x_{23} \\
s_8 &= -30 - 11x_{11} - x_{11}^2 \\
&\quad -4x_{11}x_{23} - 22x_{23} - 4x_{23}^2 \\
&\quad +x_{24} \\
s_9 &= -380 - 312x_{23} - 64x_{23}^2 \\
&\quad -32x_{23}x_{24} - 78x_{24} - 4x_{24}^2 \\
&\quad +x_{25} \\
s_{10} &= -6 - 5x_{12} - x_{12}^2 \\
&\quad -2x_{12}x_{13} - 5x_{13} - x_{13}^2 \\
&\quad +x_{26} \\
s_{11} &= -6 - 5x_{11} - x_{11}^2 \\
&\quad -2x_{11}x_{13} - 5x_{13} - x_{13}^2 \\
&\quad +x_{27} \\
s_{12} &= -414 - 22x_{11} - 2x_{11}^2 \\
&\quad -16x_{11}x_{13} - 8x_{11}x_{26} - 200x_{12} \\
&\quad -32x_{12}^2 - 64x_{12}x_{13} - 32x_{12}x_{22} \\
&\quad -32x_{12}x_{27} - 320x_{13} - 64x_{13}^2 \\
&\quad -32x_{13}x_{22} - 32x_{13}x_{26} - 32x_{13}x_{27} \\
&\quad -100x_{22} - 8x_{22}^2 - 16x_{22}x_{27} \\
&\quad -52x_{26} - 8x_{26}^2 - 108x_{27} \\
&\quad -8x_{27}^2 + x_{28} \\
s_{13} &= -1344 - 800x_{12} - 128x_{12}^2 \\
&\quad -256x_{12}x_{13} - 128x_{12}x_{22} - 128x_{12}x_{27} \\
&\quad -864x_{13} - 128x_{13}^2 - 128x_{13}x_{22} \\
&\quad -128x_{13}x_{27} - 400x_{22} - 32x_{22}^2 \\
&\quad -64x_{22}x_{27} - 432x_{27} - 32x_{27}^2 \\
&\quad -2x_{28} - 2x_{28}^2 + x_{29} \\
s_{14} &= -247 - 168x_{11} - 32x_{11}^2 \\
&\quad -64x_{11}x_{12} - 32x_{11}x_{26} - 32x_{11}x_{27} \\
&\quad -188x_{12} - 32x_{12}^2 - 32x_{12}x_{26}
\end{aligned}
$$

$$
\begin{aligned}
&-32x_{12}x_{27} - 2x_{22} - 92x_{26}\\
&-8x_{26}^2 - 16x_{26}x_{27} - 84x_{27}\\
&-8x_{27}^2 + x_{30}\\
s_{15} = &-960 - 672x_{11} - 128x_{11}^2\\
&-256x_{11}x_{12} - 128x_{11}x_{26} - 128x_{11}x_{27}\\
&-736x_{12} - 128x_{12}^2 - 128x_{12}x_{26}\\
&-128x_{12}x_{27} - 368x_{26} - 32x_{26}^2\\
&-64x_{26}x_{27} - 336x_{27} - 32x_{27}^2\\
&-2x_{30} - 2x_{30}^2 + x_{31}\\
s_{16} = &-276 - 30x_{11} - 2x_{11}^2\\
&-16x_{11}x_{13} - 8x_{11}x_{26} - 30x_{12}\\
&-2x_{12}^2 - 16x_{12}x_{13} - 8x_{12}x_{27}\\
&-272x_{13} - 64x_{13}^2 - 32x_{13}x_{26}\\
&-32x_{13}x_{27} - 68x_{26} - 8x_{26}^2\\
&-68x_{27} - 8x_{27}^2 + x_{32}\\
s_{17} = &-552 - 120x_{12} - 8x_{12}^2\\
&-64x_{12}x_{13} - 32x_{12}x_{27} - 544x_{13}\\
&-128x_{13}^2 - 128x_{13}x_{27} - 272x_{27}\\
&-32x_{27}^2 - 2x_{32} - 2x_{32}^2\\
&+x_{33}\\
s_{18} = &-247 - 4x_{11} - 352x_{12}\\
&-128x_{12}^2 - 64x_{12}x_{22} - 64x_{12}x_{26}\\
&-92x_{22} - 8x_{22}^2 - 16x_{22}x_{26}\\
&-84x_{26} - 8x_{26}^2 - 2x_{27}\\
&+x_{34}\\
s_{19} = &-960 - 1408x_{12} - 512x_{12}^2\\
&-256x_{12}x_{22} - 256x_{12}x_{26} - 368x_{22}\\
&-32x_{22}^2 - 64x_{22}x_{26} - 336x_{26}\\
&-32x_{26}^2 - 2x_{34} - 2x_{34}^2\\
&+x_{35}\\
s_{20} = &-56 - 60x_{12} - 16x_{12}^2\\
&-16x_{12}x_{22} - 30x_{22} - 4x_{22}^2\\
&+x_{36}\\
s_{21} = &-30 - 11x_{11} - x_{11}^2\\
&-4x_{11}x_{36} - 22x_{36} - 4x_{36}^2\\
&+x_{37}\\
s_{22} = &-380 - 312x_{36} - 64x_{36}^2\\
&-32x_{36}x_{37} - 78x_{37} - 4x_{37}^2\\
&+x_{38}\\
s_{23} = &-30 - 11x_{12} - x_{12}^2\\
&-4x_{12}x_{38} - 22x_{38} - 4x_{38}^2\\
&+x_{39}\\
s_{24} = &-349 - 420x_{12} - 128x_{12}^2\\
&-128x_{12}x_{22} - 208x_{22} - 32x_{22}^2\\
&-2x_{26} + x_{40}\\
s_{25} = &-1368 - 1664x_{12} - 512x_{12}^2\\
&-512x_{12}x_{22} - 832x_{22} - 128x_{22}^2\\
&-2x_{40} - 2x_{40}^2 + x_{41}\\
s_{26} = &-56 - 60x_{11} - 16x_{11}^2\\
&-16x_{11}x_{27} - 8x_{11}x_{41} - 30x_{27}\\
&-4x_{27}^2 - 4x_{27}x_{41} - 15x_{41}\\
&-x_{41}^2 + x_{42}\\
s_{27} = &-6 - 5x_{13} - x_{13}^2\\
&-2x_{13}x_{14} - 5x_{14} - x_{14}^2\\
&+x_{43}
\end{aligned}
$$

$$
\begin{aligned}
s_{28} = &-224 - 26x_{11} - 2x_{11}^2 \\
&-16x_{11}x_{14} - 8x_{11}x_{43} - 288x_{13} \\
&-128x_{13}^2 - 64x_{13}x_{26} - 64x_{13}x_{27} \\
&-88x_{14} - 32x_{14}^2 - 32x_{14}x_{43} \\
&-76x_{26} - 8x_{26}^2 - 16x_{26}x_{27} \\
&-68x_{27} - 8x_{27}^2 - 44x_{43} \\
&-8x_{43}^2 + x_{44} \\
s_{29} = &-640 - 1152x_{13} - 512x_{13}^2 \\
&-256x_{13}x_{26} - 256x_{13}x_{27} - 304x_{26} \\
&-32x_{26}^2 - 64x_{26}x_{27} - 272x_{27} \\
&-32x_{27}^2 - 2x_{44} - 2x_{44}^2 \\
&+x_{45} \\
s_{30} = &-241 - 168x_{11} - 32x_{11}^2 \\
&-64x_{11}x_{12} - 32x_{11}x_{26} - 32x_{11}x_{27} \\
&-184x_{12} - 32x_{12}^2 - 32x_{12}x_{26} \\
&-32x_{12}x_{27} - x_{13} - 92x_{26} \\
&-8x_{26}^2 - 16x_{26}x_{27} - 84x_{27} \\
&-8x_{27}^2 + x_{46} \\
s_{31} = &-42 - 52x_{13} - 16x_{13}^2 \\
&-16x_{13}x_{26} - 26x_{26} - 4x_{26}^2 \\
&+x_{47} \\
s_{32} = &-30 - 11x_{13} - x_{13}^2 \\
&-4x_{13}x_{47} - 22x_{47} - 4x_{47}^2 \\
&+x_{48} \\
s_{33} = &-306 - 35x_{12} - x_{12}^2 \\
&-16x_{12}x_{47} - 4x_{12}x_{48} - 280x_{47} \\
&-64x_{47}^2 - 32x_{47}x_{48} - 70x_{48} \\
&-4x_{48}^2 + x_{49} \\
s_{34} = &-4556 - 4320x_{47} - 1024x_{47}^2 \\
&-512x_{47}x_{48} - 128x_{47}x_{49} - 1080x_{48} \\
&-64x_{48}^2 - 32x_{48}x_{49} - 270x_{49} \\
&-4x_{49}^2 + x_{50} \\
s_{35} = &-30 - 11x_{13} - x_{13}^2 \\
&-4x_{13}x_{50} - 22x_{50} - 4x_{50}^2 \\
&+x_{51} \\
s_{36} = &-306 - 35x_{12} - x_{12}^2 \\
&-16x_{12}x_{50} - 4x_{12}x_{51} - 280x_{50} \\
&-64x_{50}^2 - 32x_{50}x_{51} - 70x_{51} \\
&-4x_{51}^2 + x_{52} \\
s_{37} = &-1028 - 672x_{11} - 128x_{11}^2 \\
&-256x_{11}x_{12} - 128x_{11}x_{26} - 128x_{11}x_{27} \\
&-736x_{12} - 128x_{12}^2 - 128x_{12}x_{26} \\
&-128x_{12}x_{27} - 368x_{26} - 32x_{26}^2 \\
&-64x_{26}x_{27} - 336x_{27} - 32x_{27}^2 \\
&-2x_{46} - 2x_{46}^2 - 32x_{50} \\
&-8x_{51} - 2x_{52} + x_{53} \\
s_{38} = &-264 - 128x_{50} - 32x_{51} \\
&-8x_{52} - 2x_{53} - 2x_{53}^2 \\
&+x_{54} \\
s_{39} = &-2 - 3x_{11} - x_{11}^2 \\
&-2x_{11}x_{54} - 3x_{54} - x_{54}^2 \\
&+x_{55} \\
s_{40} = &-12 - 28x_{54} - 16x_{54}^2 \\
&-16x_{54}x_{55} - 14x_{55} - 4x_{55}^2 \\
&+x_{56}
\end{aligned}
$$

$$
\begin{aligned}
s_{41} &= -30 - 11x_{13} - x_{13}^2 \\
&\quad -4x_{13}x_{56} - 22x_{56} - 4x_{56}^2 \\
&\quad +x_{57} \\
s_{42} &= -306 - 35x_{11} - x_{11}^2 \\
&\quad -16x_{11}x_{56} - 4x_{11}x_{57} - 280x_{56} \\
&\quad -64x_{56}^2 - 32x_{56}x_{57} - 70x_{57} \\
&\quad -4x_{57}^2 + x_{58} \\
s_{43} &= -4556 - 4320x_{56} - 1024x_{56}^2 \\
&\quad -512x_{56}x_{57} - 128x_{56}x_{58} - 1080x_{57} \\
&\quad -64x_{57}^2 - 32x_{57}x_{58} - 270x_{58} \\
&\quad -4x_{58}^2 + x_{59} \\
s_{44} &= -90 - 76x_{13} - 16x_{13}^2 \\
&\quad -16x_{13}x_{26} - 16x_{13}x_{59} - 38x_{26} \\
&\quad -4x_{26}^2 - 8x_{26}x_{59} - 38x_{59} \\
&\quad -4x_{59}^2 + x_{60} \\
s_{45} &= -4556 - 4320x_{50} - 1024x_{50}^2 \\
&\quad -512x_{50}x_{51} - 128x_{50}x_{52} - 1080x_{51} \\
&\quad -64x_{51}^2 - 32x_{51}x_{52} - 270x_{52} \\
&\quad -4x_{52}^2 + x_{61} \\
s_{46} &= -682 - 70x_{12} - 2x_{12}^2 \\
&\quad -32x_{12}x_{59} - 8x_{12}x_{60} - 592x_{59} \\
&\quad -128x_{59}^2 - 64x_{59}x_{60} - 148x_{60} \\
&\quad -8x_{60}^2 - 2x_{61} + x_{62} \\
s_{47} &= -x_{62} - x_{62}^2 + x_{63} \\
s_{48} &= -42 - 52x_{12} - 16x_{12}^2 \\
&\quad -8x_{12}x_{13} - 16x_{12}x_{22} - 13x_{13} \\
&\quad -x_{13}^2 - 4x_{13}x_{22} - 26x_{22} \\
&\quad -4x_{22}^2 + x_{64} \\
s_{49} &= -2278 - 2264x_{12} - 800x_{12}^2 \\
&\quad -800x_{12}x_{22} - 160x_{12}x_{64} - 70x_{13} \\
&\quad -2x_{13}^2 - 32x_{13}x_{61} - 8x_{13}x_{63} \\
&\quad -1132x_{22} - 200x_{22}^2 - 80x_{22}x_{64} \\
&\quad -592x_{61} - 128x_{61}^2 - 64x_{61}x_{63} \\
&\quad -148x_{63} - 8x_{63}^2 - 228x_{64} \\
&\quad -8x_{64}^2 + x_{65} \\
s_{50} &= -2712 - 280x_{13} - 8x_{13}^2 \\
&\quad -128x_{13}x_{61} - 32x_{13}x_{63} - 2368x_{61} \\
&\quad -512x_{61}^2 - 256x_{61}x_{63} - 592x_{63} \\
&\quad -32x_{63}^2 - 2x_{65} - 2x_{65}^2 \\
&\quad +x_{66} \\
s_{51} &= -2 - 3x_{66} - x_{66}^2 \\
&\quad +x_{67} \\
s_{52} &= -2 - x_2 - x_2x_{16} \\
&\quad -2x_{16} + x_{68} \\
s_{53} &= -x_{15}^2 + 2x_{15}x_{16} - x_{16}^2 \\
&\quad +x_{69} \\
s_{54} &= -2x_{25}x_{68} - 4x_{68} + x_{70} \\
s_{55} &= -x_{15}^2 + 2x_{15}x_{16} + 2x_{15}x_{70} \\
&\quad -x_{16}^2 - 2x_{16}x_{70} - x_{70}^2 \\
&\quad +x_{71} \\
s_{56} &= -x_{29}x_{68} + x_{72} \\
s_{57} &= -x_{15}^2 + 2x_{15}x_{16} + 2x_{15}x_{72} \\
&\quad -x_{16}^2 - 2x_{16}x_{72} - x_{72}^2 \\
&\quad +x_{73}
\end{aligned}
$$

$$
\begin{aligned}
s_{58} &= -x_{31}x_{68} + x_{74} \\
s_{59} &= -x_{15}^2 + 2x_{15}x_{16} + 2x_{15}x_{74} \\
&\quad -x_{16}^2 - 2x_{16}x_{74} - x_{74}^2 \\
&\quad +x_{75} \\
s_{60} &= -x_{33}x_{68} + x_{76} \\
s_{61} &= -x_{15}^2 + 2x_{15}x_{16} + 2x_{15}x_{76} \\
&\quad -x_{16}^2 - 2x_{16}x_{76} - x_{76}^2 \\
&\quad +x_{77} \\
s_{62} &= -x_{35}x_{68} + x_{78} \\
s_{63} &= -x_{15}^2 + 2x_{15}x_{16} + 2x_{15}x_{78} \\
&\quad -x_{16}^2 - 2x_{16}x_{78} - x_{78}^2 \\
&\quad +x_{79} \\
s_{64} &= -4x_{12}x_{68} - 2x_{39}x_{68} - 4x_{68} \\
&\quad +x_{80} \\
s_{65} &= -x_{15}^2 + 2x_{15}x_{16} + 2x_{15}x_{80} \\
&\quad -x_{16}^2 - 2x_{16}x_{80} - x_{80}^2 \\
&\quad +x_{81} \\
s_{66} &= -4x_{41}x_{68} - 2x_{42}x_{68} + x_{82} \\
s_{67} &= -x_{15}^2 + 2x_{15}x_{16} + 2x_{15}x_{82} \\
&\quad -x_{16}^2 - 2x_{16}x_{82} - x_{82}^2 \\
&\quad +x_{83} \\
s_{68} &= -x_{45}x_{68} + x_{84} \\
s_{69} &= -x_{15}^2 + 2x_{15}x_{16} + 2x_{15}x_{84} \\
&\quad -x_{16}^2 - 2x_{16}x_{84} - x_{84}^2 \\
&\quad +x_{85} \\
s_{70} &= -x_{11}^2 + 2x_{11}x_{13} - x_{13}^2 \\
&\quad +x_{86} \\
s_{71} &= -x_{12}^2 + 2x_{12}x_{13} - x_{13}^2 \\
&\quad +x_{87} \\
s_{72} &= -x_{86}x_{87} + x_{88} \\
s_{73} &= -4 - 4x_{14} - x_{14}^2 \\
&\quad +x_{89} \\
s_{74} &= -x_{88}^2 + 2x_{88}x_{89} - x_{89}^2 \\
&\quad +x_{90} \\
s_{75} &= -2x_{67}x_{68} - 4x_{68} + x_{91} \\
s_{76} &= -x_{15}^2 + 2x_{15}x_{16} + 2x_{15}x_{91} \\
&\quad -x_{16}^2 - 2x_{16}x_{91} - x_{91}^2 \\
&\quad +x_{92} \\
s_{77} &= -2 - x_2 - x_2x_{12} \\
&\quad -2x_{12} + x_{93} \\
s_{78} &= -1 - x_{16} - x_{16}x_{93} \\
&\quad -x_{93} + x_{94} \\
s_{79} &= -2 - x_2 - x_2x_{11} \\
&\quad -2x_{11} + x_{95} \\
s_{80} &= -8 - 4x_2 - 4x_2x_{12} \\
&\quad -2x_2x_{22} - 8x_{12} - 4x_{22} \\
&\quad +x_{96} \\
s_{81} &= -1 - x_{15} - x_{15}x_{95} \\
&\quad -x_{95} + x_{97} \\
s_{82} &= -x_{96}x_{97} - x_{97} + x_{98} \\
s_{83} &= -x_{94}^2 + 2x_{94}x_{98} - x_{98}^2 \\
&\quad +x_{99}
\end{aligned}
$$

$$
\begin{aligned}
s_{84} &= -12 - 6x_2 - 4x_2x_{12} \\
&\quad -2x_2x_{22} - 8x_{12} - 4x_{22} \\
&\quad +x_{100} \\
s_{85} &= -1 - x_{16} - x_{16}x_{100} \\
&\quad -x_{100} + x_{101} \\
s_{86} &= -1 - x_{15} - x_{15}x_{93} \\
&\quad -x_{93} + x_{102} \\
s_{87} &= -x_{101}^2 + 2x_{101}x_{102} - x_{102}^2 \\
&\quad +x_{103} \\
s_{88} &= -x_{69}x_{71} + x_{104} \\
s_{89} &= -x_{73}x_{104} + x_{105} \\
s_{90} &= -x_{75}x_{105} + x_{106} \\
s_{91} &= -x_{77}x_{106} + x_{107} \\
s_{92} &= -x_{79}x_{107} + x_{108} \\
s_{93} &= -x_{81}x_{108} + x_{109} \\
s_{94} &= -x_{83}x_{109} + x_{110} \\
s_{95} &= -x_{85}x_{110} + x_{111} \\
s_{96} &= -x_{90}x_{111} - x_{92}x_{111} + x_{112} \\
s_{97} &= -x_{99}x_{112} + x_{113} \\
s_{98} &= -6 - 2x_2 - x_2x_{19} \\
&\quad -2x_8 - x_8x_{19} - 3x_{19} \\
&\quad +x_{114} \\
s_{99} &= -1 - x_{17} - x_{17}x_{114} \\
&\quad -x_{114} + x_{115} \\
s_{100} &= -4 - x_2 - x_2x_{18} \\
&\quad -x_8 - x_8x_{18} - 4x_{18} \\
&\quad -x_{18}x_{114} - x_{114} + x_{116} \\
s_{101} &= 2 - 4x_1 + 2x_1^2 \\
&\quad -2x_1x_{15} - 2x_1x_{16} - 2x_1x_{115} \\
&\quad -2x_1x_{116} + 2x_{15} + x_{15}^2 \\
&\quad +2x_{15}x_{115} + 2x_{16} + x_{16}^2 \\
&\quad +2x_{16}x_{116} - x_{20}x_{117} + x_{103}x_{113} \\
&\quad +2x_{115} + x_{115}^2 + 2x_{116} \\
&\quad +x_{116}^2 \\
s_{102} &= -9 - x_{11} - x_{12} \\
&\quad -x_{13} - x_{14} - x_{15} \\
&\quad -x_{16} - x_{17} - x_{18} \\
&\quad -x_{19} + 10x_{20}^2 - x_{118} \\
s_{103} &= -2 - x_4 - x_4x_{169} \\
&\quad -x_4x_{170} - 3x_{169} - x_{169}x_{171} \\
&\quad -3x_{170} - x_{170}x_{171} - x_{171} \\
&\quad +x_{176} + x_{177} \\
s_{104} &= -x_{119} - x_{119}x_{122} + x_{123} \\
s_{105} &= -x_{119}x_{123} - x_{123} + x_{124} \\
s_{106} &= -1 + x_{120}^2 + x_{123}^2 \\
&\quad -x_{124}^2 \\
s_{107} &= -17 - 8x_4 - x_4^2 \\
&\quad +2x_4x_{120} + 2x_4x_{121} - 2x_4x_{171} \\
&\quad +8x_{120} + 2x_{120}x_{171} + 8x_{121} \\
&\quad +2x_{121}x_{171} - x_{125}x_{126} - 8x_{171} \\
&\quad -x_{171}^2 \\
s_{108} &= -x_{20} - x_{20}x_{214} - x_{20}x_{215}
\end{aligned}
$$

$$
\begin{aligned}
&\quad +x_{126}\\
s_{109} &= -4x_{128} - 4x_{128}x_{133} + x_{134}\\
s_{110} &= -x_{120}x_{128} - x_{121}x_{128} + x_{135}\\
s_{111} &= -1 + x_{127}^2 + x_{128}^2\\
&\quad -x_{135}^2\\
s_{112} &= -x_{120}x_{130} - x_{121}x_{130} + x_{136}\\
s_{113} &= -1 + x_{129}^2 + x_{130}^2\\
&\quad -x_{136}^2\\
s_{114} &= -x_{132} - x_{132}x_{134} + x_{137}\\
s_{115} &= -1 + x_{131}^2 + x_{132}^2\\
&\quad -x_{137}^2\\
s_{116} &= -x_{128}^2 + x_{139}\\
s_{117} &= x_{130} - x_{138}x_{139} - x_{139}\\
s_{118} &= 1 - x_{120} - x_{121}\\
&\quad -x_{129}x_{140} + x_{134}\\
s_{119} &= -x_{127} - x_{129}x_{141} + x_{131}\\
s_{120} &= -25 - x_{11} - 13x_{12}\\
&\quad -4x_{22} - 2x_{25} - x_{29}\\
&\quad -x_{31} - x_{33} - x_{35}\\
&\quad -2x_{39} - 4x_{41} - 2x_{42}\\
&\quad -x_{45} - 2x_{67} - 4x_{128}x_{142}\\
&\quad +x_{132}\\
s_{121} &= -x_{143}x_{144} + x_{145}\\
s_{122} &= -1 - 2x_{120}x_{145} - 2x_{121}x_{145}\\
&\quad +x_{143}^2 + x_{144}^2\\
s_{123} &= -1 - x_3^2 + 2x_3x_{120}\\
&\quad +2x_3x_{121} - x_{146}\\
s_{124} &= -4 - 2x_2 - x_2x_4\\
&\quad +x_3x_{143} - 2x_4 - x_5\\
&\quad -x_{144} - x_{146}x_{147}\\
s_{125} &= -x_{148}x_{149} + x_{150}\\
s_{126} &= -1 - 2x_{120}x_{150} - 2x_{121}x_{150}\\
&\quad +x_{148}^2 + x_{149}^2\\
s_{127} &= -5 - 4x_4 - x_4^2\\
&\quad +2x_4x_{120} + 2x_4x_{121} + 4x_{120}\\
&\quad +4x_{121} - x_{151}\\
s_{128} &= x_4x_{148} - x_{20} + 2x_{148}\\
&\quad -x_{149} - x_{151}x_{152}\\
s_{129} &= -1 - x_{20}^2 + 2x_{20}x_{120}\\
&\quad +2x_{20}x_{121} - x_{153}\\
s_{130} &= -3 - x_4 + x_{20}x_{148}\\
&\quad -x_{149} - x_{153}x_{154} - x_{171}\\
s_{131} &= -10 - 6x_4 - x_4^2\\
&\quad +2x_4x_{120} + 2x_4x_{121} - 2x_4x_{171}\\
&\quad +6x_{120} + 2x_{120}x_{171} + 6x_{121}\\
&\quad +2x_{121}x_{171} - x_{155} - 6x_{171}\\
&\quad -x_{171}^2\\
s_{132} &= -1 + x_4x_{143} + 3x_{143}\\
&\quad +x_{143}x_{171} - x_{144} - x_{155}x_{156}\\
&\quad -x_{169} - x_{170}\\
s_{133} &= -x_{157}x_{158} + x_{159}\\
s_{134} &= -1 - 2x_{120}x_{159} - 2x_{121}x_{159}
\end{aligned}
$$

$$
\begin{aligned}
&+x_{157}^2 + x_{158}^2 \\
s_{135} = &-1 + x_4 x_{157} - x_{155} x_{160} \\
&+3x_{157} + x_{157} x_{171} - x_{158} \\
&-x_{201} - x_{202} \\
s_{136} = &-x_{161} x_{162} + x_{163} \\
s_{137} = &-1 - 2x_{120} x_{163} - 2x_{121} x_{163} \\
&+x_{161}^2 + x_{162}^2 \\
s_{138} = &-1 + x_4 x_{161} - x_{155} x_{164} \\
&+3x_{161} + x_{161} x_{171} - x_{162} \\
&-x_{214} - x_{215} \\
s_{139} = &-x_{165} x_{166} + x_{167} \\
s_{140} = &-1 - 2x_{120} x_{167} - 2x_{121} x_{167} \\
&+x_{165}^2 + x_{166}^2 \\
s_{141} = &-2 - x_4 + x_4 x_{165} \\
&-x_4 x_{169} - x_4 x_{170} + 4x_{165} \\
&+x_{165} x_{171} - x_{166} - x_{168} \\
&-x_{168} x_{176} - x_{168} x_{177} - 3x_{169} \\
&-2x_{170} \\
s_{142} = &-x_{172} x_{173} + x_{174} \\
s_{143} = &-1 - 2x_{120} x_{174} - 2x_{121} x_{174} \\
&+x_{172}^2 + x_{173}^2 \\
s_{144} = &-x_{20} - x_{20} x_{176} - x_{20} x_{177} \\
&+x_{178} \\
s_{145} = &\, x_4 x_{172} + x_{171} x_{172} + 4x_{172} \\
&-x_{173} - x_{175} x_{178} - x_{176} \\
s_{146} = &-x_{179} x_{180} + x_{181} \\
s_{147} = &-1 - 2x_{120} x_{181} - 2x_{121} x_{181} \\
&+x_{179}^2 + x_{180}^2 \\
s_{148} = &\, x_4 x_{179} + x_{171} x_{179} - x_{178} x_{182} \\
&+4x_{179} - x_{180} - x_{183} \\
s_{149} = &\, x_{176} + x_{177} - x_{183} \\
&-x_{184} \\
s_{150} = &-x_{185} x_{186} + x_{187} \\
s_{151} = &-1 - 2x_{120} x_{187} - 2x_{121} x_{187} \\
&+x_{185}^2 + x_{186}^2 \\
s_{152} = &\, x_4 x_{185} + x_{171} x_{185} - x_{178} x_{188} \\
&+4x_{185} - x_{186} - x_{189} \\
s_{153} = &\, x_{176} + x_{177} - x_{189} \\
&-x_{190} \\
s_{154} = &-x_{191} x_{192} + x_{193} \\
s_{155} = &-1 - 2x_{120} x_{193} - 2x_{121} x_{193} \\
&+x_{191}^2 + x_{192}^2 \\
s_{156} = &\, x_4 x_{191} + x_{171} x_{191} - x_{178} x_{194} \\
&+4x_{191} - x_{192} - x_{195} \\
s_{157} = &\, x_{176} + x_{177} - x_{195} \\
&-x_{196} \\
s_{158} = &-x_{197} x_{198} + x_{199} \\
s_{159} = &-1 - 2x_{120} x_{199} - 2x_{121} x_{199} \\
&+x_{197}^2 + x_{198}^2 \\
s_{160} = &-x_{20} - x_{20} x_{201} - x_{20} x_{202} \\
&+x_{203} \\
s_{161} = &\, x_4 x_{197} + x_{171} x_{197} + 4x_{197}
\end{aligned}
$$

$$
\begin{aligned}
&\quad -x_{198} - x_{200}x_{203} - x_{201} \\
s_{162} &= -x_{204}x_{205} + x_{206} \\
s_{163} &= -1 - 2x_{120}x_{206} - 2x_{121}x_{206} \\
&\quad +x_{204}^2 + x_{205}^2 \\
s_{164} &= x_4x_{204} + x_{171}x_{204} - x_{203}x_{207} \\
&\quad +4x_{204} - x_{205} - x_{208} \\
s_{165} &= x_{201} + x_{202} - x_{208} \\
&\quad -x_{209} \\
s_{166} &= -x_{210}x_{211} + x_{212} \\
s_{167} &= -1 - 2x_{120}x_{212} - 2x_{121}x_{212} \\
&\quad +x_{210}^2 + x_{211}^2 \\
s_{168} &= x_4x_{210} - x_{126}x_{213} + x_{171}x_{210} \\
&\quad +4x_{210} - x_{211} - x_{214} \\
s_{169} &= -x_{216}x_{217} + x_{218} \\
s_{170} &= -1 - 2x_{120}x_{218} - 2x_{121}x_{218} \\
&\quad +x_{216}^2 + x_{217}^2 \\
s_{171} &= x_4x_{216} - x_{126}x_{219} + x_{171}x_{216} \\
&\quad +4x_{216} - x_{217} - x_{220} \\
s_{172} &= x_{214} + x_{215} - x_{220} \\
&\quad -x_{221} \\
s_{173} &= -x_{222}x_{223} + x_{224} \\
s_{174} &= -1 - 2x_{120}x_{224} - 2x_{121}x_{224} \\
&\quad +x_{222}^2 + x_{223}^2 \\
s_{175} &= -x_{20} - x_{20}x_{169} - x_{20}x_{170} \\
&\quad +x_{228} \\
s_{176} &= x_4x_{222} + x_{171}x_{222} + 4x_{222} \\
&\quad -x_{223} - x_{225}x_{228} - x_{226} \\
s_{177} &= x_{169} + x_{170} - x_{226} \\
&\quad -x_{227} \\
s_{178} &= x_4x_{128} + x_{127} + 4x_{128} \\
&\quad +x_{128}x_{171} - x_{135} - x_{178}x_{229} \\
&\quad -x_{230} \\
s_{179} &= x_{176} + x_{177} - x_{230} \\
&\quad -x_{231} \\
s_{180} &= -1639 - 176x_0 - 5x_0^2 \\
&\quad -8x_0x_1 - 16x_0x_2 - 8x_0x_8 \\
&\quad +2x_0x_{143} + 4x_0x_{148} - 154x_1 \\
&\quad -5x_1^2 - 16x_1x_2 - 8x_1x_8 \\
&\quad +4x_1x_{148} + 2x_1x_{161} - 312x_2 \\
&\quad -17x_2^2 - 18x_2x_8 + 8x_2x_{148} \\
&\quad +2x_2x_{157} - x_3^2 + 2x_3x_{165} \\
&\quad -160x_8 - 5x_8^2 + 4x_8x_{148} \\
&\quad +2x_8x_{157} - 3x_{11} - x_{11}^2 \\
&\quad +2x_{11}x_{172} - 15x_{12} - x_{12}^2 \\
&\quad +2x_{12}x_{179} - 2x_{13} - x_{13}^2 \\
&\quad +2x_{13}x_{185} - 2x_{14} - x_{14}^2 \\
&\quad +2x_{14}x_{191} - 2x_{15} - x_{15}^2 \\
&\quad +2x_{15}x_{197} - 2x_{16} - x_{16}^2 \\
&\quad +2x_{16}x_{204} - 2x_{17} - x_{17}^2 \\
&\quad +2x_{17}x_{210} - 2x_{18} - x_{18}^2 \\
&\quad +2x_{18}x_{216} - 2x_{19} - x_{19}^2 \\
&\quad +2x_{19}x_{222} - 4x_{22} - 2x_{25} \\
&\quad -x_{29} - x_{31} - x_{33} \\
&\quad -x_{35} - 2x_{39} - 4x_{41}
\end{aligned}
$$

$$
\begin{aligned}
&-2x_{42} - x_{45} - 2x_{67} \\
&-x_{120}x_{232} - x_{121}x_{232} + x_{128} \\
&+24x_{143} - x_{143}^2 + 76x_{148} \\
&-x_{148}^2 + 8x_{157} - x_{157}^2 \\
&+2x_{161} - x_{161}^2 - x_{165}^2 \\
&+2x_{172} - x_{172}^2 + 2x_{179} \\
&-x_{179}^2 + 2x_{185} - x_{185}^2 \\
&+2x_{191} - x_{191}^2 + 2x_{197} \\
&-x_{197}^2 + 2x_{204} - x_{204}^2 \\
&+2x_{210} - x_{210}^2 + 2x_{216} \\
&-x_{216}^2 + 2x_{222} - x_{222}^2 \\
&+x_{232}
\end{aligned}
$$

$$
\begin{aligned}
s_{181} = &-117 - 3x_0 - 3x_1 \\
&-7x_2 - x_2x_4 - x_3 \\
&-4x_4 - x_5 - 3x_8 \\
&-2x_{11} - 14x_{12} - x_{13} \\
&-x_{14} - x_{15} - x_{16} \\
&-x_{17} - x_{18} - x_{19} \\
&-x_{20} - 4x_{22} - 2x_{25} \\
&-x_{29} - x_{31} - x_{33} \\
&-x_{35} - 2x_{39} - 4x_{41} \\
&-2x_{42} - x_{45} - 2x_{67} \\
&+x_{119} - x_{169} - x_{170} \\
&-x_{171} - x_{201} - x_{202} \\
&-x_{214} - x_{215}
\end{aligned}
$$

C The four fixed compact coefficients

The recipe below is reproduced from the pinned indexed native source. It supplies all four constants in Section 4; the symbol K in this recipe is its K_0. All bounds and indices here are fixed integers, not additional unknowns. The code in `compact/rebuild_coefficients.py` independently evaluates the recipe and checks its four literal payloads. **The four fixed integers.** Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$
\begin{aligned}
\mathcal{E} = &\{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\
&\cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.
\end{aligned}
$$

Then

$$
\begin{aligned}
M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\
M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\
&\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\
&\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\
K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\
&\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),
\end{aligned}
$$

where

$$\Phi(x) = (1+2^{51})G_{943}(x) - 2^{10}\sum_{i=0}^{942} v(t_i)x^i + \sum_{(b,e,m)\in\mathcal{C}} 2^b x^e G_m(2x),$$
$$\mathcal{C} = \{(10,1023,8),(18,1013,9),(27,1003,8),(35,983,9),(44,1031,1),(45,1022,1),(46,1011,1),(47,992,1),(48,1095,1),(49,2195,1),(50,3296,1)\}.$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$h(q,b) = 2(q+1)+b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\},$$
$$\mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\},$$
$$r_q = 6 + |\mathcal{R} \cap \{0,\dots,q-1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0,\dots,q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w,n,e,s)$
$0, 1$	$(2,435,3,1),\ (5,0,0,0)$
$j+2\ (j = 2q+b < 430)$	$((1-d_j)l_{u_j}, h(q,b), d_j r_{u_j}, v_j)$
$432+b\ (b=0,1)$	$(0, h(215,b), 0, h(215,b))$
$434+b\ (b=0,1)$	$(0,b,0,b)$
$424+2r_q+b\ (q\in\mathcal{R}, b=0,1)$	$(r_q,b,0,h(q,b))$
$424+2l_q+b\ (q\in\mathcal{L}, b=0,1)$	$(0,b,l_q,h(q,b))$
$924+j\ (0\le j<4)$	$(0, j+430\lfloor j/2\rfloor, 0, 438)$
928	$(2,434,2,0)$
929	$(3,436,4,4)$
930	$(4,436,4,0)$
$497+t\ (t=434,435,436)$	$(1,438,1,t)$
$934+3a+b-6\lfloor a/4\rfloor$ $a\in\{0,1,4\},\ b=0,1,2$	$(a,437,b,437)$

For $t=(w,n,e,s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\,2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A,k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b\rfloor$. For $0 \le q < 215$, $b=0,1$, $j = 2q+b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N,j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430+q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1-m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1+\sigma_j) + (1-\nu_j)D_{216}(T_0, j-\sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b+1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSD0Tgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```